Engineering Mathematics

Detailed Solutions

GATE ENGINEERING MATHEMATICS

PREPFUSION

Previous Year Questions Solutions Series

Contents

SOLUTIONS — UNIT

I

Vector Calculus

Topics

1. Basics of Vector Algebra
2. Cylindrical and Spherical Coordinate Systems
3. Gradient, Divergence and Curl
4. Line Integrals and Surface Integrals
5. Green's Theorem
6. Stoke's Theorem and Gauss Divergence Theorem

Vector Calculus

Engineering Mathematics

Q4.26.1 MCQ 2026 2M Ans: D ● ○ ○

Given surface:

$$z^2 = 2x^2 - y^2$$

Define

$$F(x, y, z) = 2x^2 - y^2 - z^2$$

Then the normal vector to the surface is given by

$$\nabla F = \left(\frac{\partial F}{\partial x}\right)\hat{i} + \left(\frac{\partial F}{\partial y}\right)\hat{j} + \left(\frac{\partial F}{\partial z}\right)\hat{k}$$

$$\nabla F = 4x\hat{i} - 2y\hat{j} - 2z\hat{k}$$

The point is

$$\vec{P} = \hat{i} + \sqrt{2}\hat{k}$$

Hence,

$$x = 1, \qquad y = 0, \qquad z = \sqrt{2}$$

Substituting:

$$\nabla F = 4\hat{i} - 2\sqrt{2}\hat{k}$$

Magnitude:

$$|\nabla F| = \sqrt{4^2 + (2\sqrt{2})^2} = \sqrt{16 + 8} = 2\sqrt{6}$$

Therefore unit normal vector is

$$\hat{n} = \frac{4\hat{i} - 2\sqrt{2}\hat{k}}{2\sqrt{6}} = \frac{2\hat{i} - \sqrt{2}\hat{k}}{\sqrt{6}}$$

Rationalizing:

$$\hat{n} = \frac{\sqrt{2}\hat{i} - \hat{k}}{\sqrt{3}}$$

$$\boxed{\text{(d)}\ \frac{\sqrt{2}\hat{i} - \hat{k}}{\sqrt{3}}}$$

Q4.24.1 MSQ 2024 1M Ans: A,C ● ○ ○

Given equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho u) = 0$$

Rearranging,

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot (\rho u)$$

Integrating over the arbitrary volume V,

$$\int_V \frac{\partial \rho}{\partial t}\, dv = -\int_V \nabla \cdot (\rho u)\, dv$$

Hence, option C is correct.
Using the divergence theorem,

$$\int_V \nabla \cdot (\rho u)\, dv = \oint_S \rho u \cdot \hat{n}\, ds$$

Therefore,

$$\int_V \frac{\partial \rho}{\partial t}\, dv = -\oint_S \rho u \cdot \hat{n}\, ds$$

Hence, option A is also correct.

$$\boxed{\text{A, C}}$$

Q4.24.2 MSQ | 2024 | 2M | Ans: A,B,D | ● ○ ○

Given that for every pair of points A and B, the line integral

$$\int_C (F_1\,dx + F_2\,dy + F_3\,dz)$$

is independent of path, the vector field is conservative.
Define

$$\vec{F} = F_1\hat{i} + F_2\hat{j} + F_3\hat{k}$$

For a conservative vector field,

$$\vec{F} = \nabla f$$

for some scalar potential function $f(x, y, z)$.
Also,

$$\nabla \times \vec{F} = 0$$

Checking the options one by one:
Option A:
For any conservative field, line integral over a closed path is zero:

$$\oint_\Gamma \vec{F} \cdot d\vec{r} = 0$$

Hence,

$$\oint_\Gamma (F_1\,dx + F_2\,dy + F_3\,dz) = 0$$

Therefore, option A is correct.
Option B:
Since the field is conservative, there exists a scalar potential function $f(x, y, z)$ such that

$$F_1 = \frac{\partial f}{\partial x}, \qquad F_2 = \frac{\partial f}{\partial y}, \qquad F_3 = \frac{\partial f}{\partial z}$$

Therefore, option B is correct.
Option C:
Option C states:

$$\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} = 0$$

This is the condition

$$\nabla \cdot \vec{F} = 0$$

A conservative field need not have zero divergence.
Therefore, option C is incorrect.
Option D:
Since

$$\nabla \times \vec{F} = 0$$

we get

$$\frac{\partial F_3}{\partial y} = \frac{\partial F_2}{\partial z}, \qquad \frac{\partial F_1}{\partial z} = \frac{\partial F_3}{\partial x}, \qquad \frac{\partial F_2}{\partial x} = \frac{\partial F_1}{\partial y}$$

Therefore, option D is correct.

$$\boxed{A, B, D}$$

Q4.23.1 MCQ 2023 1M Ans: C ● ○ ○

Method 1

Given

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \qquad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

We have

$$\mathbf{v}_1 = \alpha \mathbf{v}_2 + \mathbf{e}$$

where $\mathbf{e}$ is the error vector.
Thus,

$$\mathbf{e} = \mathbf{v}_1 - \alpha \mathbf{v}_2$$

To minimize the length of $\mathbf{e}$, the error vector must be orthogonal to $\mathbf{v}_2$:

$$\mathbf{v}_2 \cdot \mathbf{e} = 0$$

Substituting,

$$\mathbf{v}_2 \cdot (\mathbf{v}_1 - \alpha \mathbf{v}_2) = 0$$

$$\mathbf{v}_2 \cdot \mathbf{v}_1 - \alpha(\mathbf{v}_2 \cdot \mathbf{v}_2) = 0$$

Hence,

$$\alpha = \frac{\mathbf{v}_2 \cdot \mathbf{v}_1}{\mathbf{v}_2 \cdot \mathbf{v}_2}$$

Now,

$$\mathbf{v}_2 \cdot \mathbf{v}_1 = (2)(1) + (1)(2) + (3)(0) = 4$$

and

$$\mathbf{v}_2 \cdot \mathbf{v}_2 = 2^2 + 1^2 + 3^2 = 14$$

Therefore,

$$\alpha = \frac{4}{14} = \frac{2}{7}$$

Method 2

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \qquad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

Given,

$$\mathbf{v}_1 = \alpha \mathbf{v}_2 + \mathbf{e}$$

where $\mathbf{e}$ is the error vector.
Thus,

$$\mathbf{e} = \mathbf{v}_1 - \alpha \mathbf{v}_2$$

Now,

$$\mathbf{e} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} - \alpha \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 - 2\alpha \\ 2 - \alpha \\ -3\alpha \end{bmatrix}$$

Hence,

$$\|\mathbf{e}\|^2 = (1 - 2\alpha)^2 + (2 - \alpha)^2 + (-3\alpha)^2$$

$$= (1 - 4\alpha + 4\alpha^2) + (4 - 4\alpha + \alpha^2) + 9\alpha^2$$

$$= 14\alpha^2 - 8\alpha + 5$$

To minimize the length of **e**, minimize

$$f(\alpha) = 14\alpha^2 - 8\alpha + 5$$

Differentiating,

$$f'(\alpha) = 28\alpha - 8$$

For minima,

$$28\alpha - 8 = 0$$

$$28\alpha = 8$$

$$\alpha = \frac{8}{28} = \frac{2}{7}$$

$$\boxed{(c)\ \frac{2}{7}}$$

Q4.23.2 MCQ 2023 1M Ans: B ● ○ ○

Given scalar field

$$f(x, y, z) = xyz$$

The rate of increase in the direction of vector

$$\mathbf{v} = (2, 1, 2)$$

at the point $(0, 2, 1)$ is the directional derivative:

$$D_{\hat{v}} f = \nabla f \cdot \hat{v}$$

First compute the gradient:

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right)$$

$$\nabla f = (yz,\ xz,\ xy)$$

At the point $(0, 2, 1)$,

$$\nabla f = (2,\ 0,\ 0)$$

Now find the unit vector along **v**:

$$|\mathbf{v}| = \sqrt{2^2 + 1^2 + 2^2} = \sqrt{9} = 3$$

Hence,

$$\hat{v} = \left(\frac{2}{3}, \frac{1}{3}, \frac{2}{3}\right)$$

Therefore,

$$D_{\hat{v}} f = (2, 0, 0) \cdot \left(\frac{2}{3}, \frac{1}{3}, \frac{2}{3}\right)$$

$$D_{\hat{v}} f = \frac{4}{3}$$

$$\boxed{(b)\ \frac{4}{3}}$$

Q4.23.3 MCQ 2023 2M Ans: B ● ○ ○

Given line integral

$$\int_P^Q \left(z^2\,dx + 3y^2\,dy + 2xz\,dz\right)$$

where

$$P(1,1,2), \qquad Q(2,3,1)$$

Let

$$A = z^2, \qquad B = 3y^2, \qquad C = 2xz$$

We check whether the field is conservative.

$$\frac{\partial A}{\partial y} = 0, \qquad \frac{\partial B}{\partial x} = 0$$

$$\frac{\partial A}{\partial z} = 2z, \qquad \frac{\partial C}{\partial x} = 2z$$

$$\frac{\partial B}{\partial z} = 0, \qquad \frac{\partial C}{\partial y} = 0$$

Hence,

$$\nabla \times \vec{F} = 0$$

and the field is conservative.
Now find the potential function ϕ.
Since

$$\frac{\partial \phi}{\partial x} = z^2$$

integrating with respect to x,

$$\phi = xz^2 + f(y,z)$$

Using

$$\frac{\partial \phi}{\partial y} = 3y^2$$

we get

$$\frac{\partial f}{\partial y} = 3y^2$$

Integrating,

$$f(y,z) = y^3 + g(z)$$

Therefore,

$$\phi = xz^2 + y^3 + g(z)$$

Using

$$\frac{\partial \phi}{\partial z} = 2xz$$

we get

$$2xz + g'(z) = 2xz$$

Hence,

$$g'(z) = 0$$

so

$$g(z) = C$$

Thus,

$$\phi = xz^2 + y^3$$

Therefore,

$$\int_P^Q \left(z^2\,dx + 3y^2\,dy + 2xz\,dz\right) = \phi(Q) - \phi(P)$$

Now,

$$\phi(2,3,1) = 2(1)^2 + 3^3 = 2 + 27 = 29$$

and

$$\phi(1,1,2) = 1(2)^2 + 1^3 = 4 + 1 = 5$$

Hence,

$$29 - 5 = 24$$

$$\boxed{\text{(b) } 24}$$

Key Points

- A vector field is conservative if:

$$\nabla \times \vec{F} = 0$$

- For conservative fields:

$$\int_P^Q \vec{F} \cdot d\vec{r} = \phi(Q) - \phi(P)$$

- Potential function is obtained by integrating component functions systematically.

Mistakes to be avoided

- Verify all curl conditions before declaring the field conservative.
- Do not forget the arbitrary functions while integrating partially.
- Evaluate the potential function carefully at both end points.

Q4.22.1 | MCQ | 2022 | 1M | Ans: A | ● ○ ○

Given vector field

$$\vec{F}(x,y) = x\vec{i} + y\vec{j}$$

The required line integral is

$$\oint_C \vec{F} \cdot (dx\,\vec{i} + dy\,\vec{j}) = \oint_C (x\,dx + y\,dy)$$

Using Green's theorem,

$$\oint_C (P\,dx + Q\,dy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$$

Here,

$$P = x, \qquad Q = y$$

Therefore,

$$\frac{\partial Q}{\partial x} = 0, \qquad \frac{\partial P}{\partial y} = 0$$

Hence,

$$\oint_C (x\,dx + y\,dy) = \iint_R (0 - 0)\,dA = 0$$

$$\boxed{\text{(a) } 0}$$

Q4.21.1 MCQ 2021 1M Ans: A ● ● ○

Given vector field,

$$\vec{F}(r) = -x\,\hat{i} + y\,\hat{j}$$

The line integral is

$$\int_C \vec{F}(r)\cdot d\vec{r} = \int_C (-x\,dx + y\,dy)$$

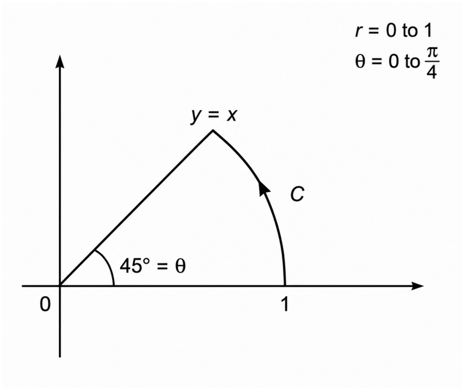

Fig. Solution Q4.21.1

From the figure, the curve C is a circular arc of radius 1 with

$$r = 0 \text{ to } 1, \qquad \theta = 0 \text{ to } \frac{\pi}{4}$$

In polar form,

$$x = \cos\theta, \qquad y = \sin\theta$$

$$dx = -\sin\theta\,d\theta, \qquad dy = \cos\theta\,d\theta$$

Substituting,

$$\int_C \vec{F}(r)\cdot d\vec{r} = \int_0^{\pi/4} [-\cos\theta(-\sin\theta\,d\theta) + \sin\theta(\cos\theta\,d\theta)]$$

$$= \int_0^{\pi/4} \sin 2\theta\,d\theta$$

$$= -\frac{\cos 2\theta}{2}\bigg|_0^{\pi/4}$$

$$= -\frac{1}{2}[0-1] = \frac{1}{2}$$

Therefore, option A is correct.

$$\boxed{\text{(a) } \frac{1}{2}}$$

Key Points

- Use polar coordinates for circular arc integration.
- Convert both coordinates and differentials into parameter form.
- Apply limits according to angular traversal of the arc.

Q4.20.1 MCQ 2020 1M Ans: C ● ○ ○

We check each option one by one.
Option A:
A vector field is said to be solenoidal if

$$\nabla \cdot \vec{A} = 0$$

Hence, option A is correct.
Option B:
Curl of a vector field is another vector field:

$$\nabla \times \vec{A}$$

Hence, option B is correct.
Option C:
A vector field is irrotational if

$$\nabla \times \vec{A} = 0$$

The condition

$$\nabla^2 \vec{A} = 0$$

represents Laplace's equation and does not imply irrotationality.
Hence, option C is false.
Option D:
Using the standard vector identity,

$$\nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}$$

Hence, option D is correct.

$$\boxed{\text{(c)}\vec{A} \text{ is irrotational if } \nabla^2 \vec{A} = 0}$$

Q4.19.1 MCQ 2019 2M Ans: C ● ○ ○

Given line integral

$$\oint_C (x\,dy - y\,dx)$$

where C is traversed in the counterclockwise direction.
Using Green's theorem,

$$\oint_C (P\,dx + Q\,dy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

Here,

$$P = -y, \qquad Q = x$$

Therefore,

$$\frac{\partial Q}{\partial x} = 1, \qquad \frac{\partial P}{\partial y} = -1$$

Hence,

$$\oint_C (x\,dy - y\,dx) = \iint_R (1 - (-1))\,dA$$

$$= 2 \iint_R dA = 2(\text{Area of } R)$$

The region R consists of a rectangle of dimensions 2×3, a semicircle of radius 1.
Area of rectangle:

$$= 2 \times 3 = 6$$

Area of semicircle:

$$= \frac{1}{2}\pi(1)^2 = \frac{\pi}{2}$$

Total area= $6 + \frac{\pi}{2}$

Therefore,

$$\oint_C (x\,dy - y\,dx) = 2\left(6 + \frac{\pi}{2}\right)$$

$$= 12 + \pi$$

$$\boxed{\text{(c) } 12 + \pi}$$

Q4.17.1 NAT 2017 2M Ans: -11 ● ○ ○

Given,

$$I = \int_C (2z\,dx + 2y\,dy + 2x\,dz)$$

The straight line segment is from $A(0, 2, 1)$ to $B(4, 1, -1)$
Parameterize the line as

$$x = 4t, \qquad y = 2 - t, \qquad z = 1 - 2t$$

where $0 \le t \le 1$.
Hence,

$$dx = 4\,dt, \qquad dy = -dt, \qquad dz = -2\,dt$$

Substituting into the integral,

$$I = \int_0^1 [2(1-2t)(4) + 2(2-t)(-1) + 2(4t)(-2)]\,dt$$

$$I = \int_0^1 (8 - 16t - 4 + 2t - 16t)\,dt$$

$$I = \int_0^1 (4 - 30t)\,dt$$

$$I = \left[4t - 15t^2\right]_0^1$$

$$I = 4 - 15 = -11$$

$$\boxed{-11}$$

Q4.17.2 MCQ 2017 2M Ans: B ● ○ ○

Given vector field,

$$\vec{F} = \hat{a}_x(3y - k_1 z) + \hat{a}_y(k_2 x - 2z) - \hat{a}_z(k_3 y + z)$$

For an irrotational vector field,

$$\nabla \times \vec{F} = 0$$

Using curl formula,

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ 3y - k_1 z & k_2 x - 2z & -(k_3 y + z) \end{vmatrix}$$

Equating each component to zero,

$$\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} = (-k_3) - (-2) = 0$$

$$k_3 = 2$$

$$\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} = (-k_1) - 0 = 0$$

$$k_1 = 0$$

$$\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} = k_2 - 3 = 0$$

$$k_2 = 3$$

Hence,

$$k_1 = 0, \qquad k_2 = 3, \qquad k_3 = 2$$

Therefore, option B is correct.

$$\boxed{\text{(b) 0.0, 3.0, 2.0}}$$

Q4.17.3 NAT 2017 1M Ans: 54.74 ● ○ ○

The angle between two planes is equal to the angle between their normal vectors.
For the planes,

$$x + y + z = 1$$

and

$$2x - y + 2z = 0$$

their normal vectors are

$$\vec{n}_1 = \hat{i} + \hat{j} + \hat{k}$$

$$\vec{n}_2 = 2\hat{i} - \hat{j} + 2\hat{k}$$

Using dot product formula,

$$\cos\theta = \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1||\vec{n}_2|}$$

$$\cos\theta = \frac{(1)(2) + (1)(-1) + (1)(2)}{\sqrt{1^2 + 1^2 + 1^2}\sqrt{2^2 + (-1)^2 + 2^2}}$$

$$\cos\theta = \frac{3}{\sqrt{3} \times 3} = \frac{1}{\sqrt{3}}$$

$$\theta = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$\theta \approx 54.74^\circ$$

$$\boxed{54.74}$$

Q4.16.1 NAT 2016 2M Ans: 20 ● ● ○

Given

$$\frac{1}{2\pi}\iint_D (x + y + 10)\, dx\, dy$$

where

$$D : x^2 + y^2 \leq 4$$

Using polar coordinates,

$$x = r\cos\theta, \qquad y = r\sin\theta$$

and

$$dx\, dy = r\, dr\, d\theta$$

Since

$$x^2 + y^2 \leq 4$$

the limits are

$$0 \leq r \leq 2, \qquad 0 \leq \theta \leq 2\pi$$

Therefore,

$$\frac{1}{2\pi}\int_0^{2\pi}\int_0^2 (r\cos\theta + r\sin\theta + 10)\, r\, dr\, d\theta$$

$$= \frac{1}{2\pi}\int_0^{2\pi}\int_0^2 (r^2\cos\theta + r^2\sin\theta + 10r)\, dr\, d\theta$$

Integrating with respect to r,

$$= \frac{1}{2\pi}\int_0^{2\pi}\left[\frac{r^3}{3}\cos\theta + \frac{r^3}{3}\sin\theta + 5r^2\right]_0^2 d\theta$$

$$= \frac{1}{2\pi}\int_0^{2\pi}\left(\frac{8}{3}\cos\theta + \frac{8}{3}\sin\theta + 20\right) d\theta$$

Now, $\int_0^{2\pi}\cos\theta\, d\theta = 0$ and $\int_0^{2\pi}\sin\theta\, d\theta = 0$
Hence, above integral reduces to

$$\frac{1}{2\pi}\int_0^{2\pi} 20\, d\theta$$

$$= \frac{1}{2\pi}(20)(2\pi)$$

$$= 20$$

$$\boxed{20}$$

Q4.16.2 NAT 2016 2M Ans: 0 ● ○ ○

Given

$$\oint_C (xy^2\,dx + x^2y\,dy)$$

where C is the circle

$$x^2 + y^2 = 1$$

oriented anti-clockwise.
Using Green's theorem,

$$\oint_C (P\,dx + Q\,dy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy$$

Here,

$$P = xy^2, \qquad Q = x^2y$$

Therefore,

$$\frac{\partial Q}{\partial x} = 2xy$$

and

$$\frac{\partial P}{\partial y} = 2xy$$

Hence,

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2xy - 2xy = 0$$

Thus,

$$\oint_C (xy^2\,dx + x^2y\,dy) = \iint_R 0\,dx\,dy = 0$$

$$\boxed{0}$$

Q4.15.1 MCQ 2015 2M Ans: A ● ○ ○

Given,

$$\vec{P} = x^3y\,\hat{a}_x - x^2y^2\,\hat{a}_y - x^2yz\,\hat{a}_z$$

To check whether the vector is solenoidal or irrotational, calculate divergence and curl.

$$\nabla\cdot\vec{P} = \frac{\partial}{\partial x}(x^3y) + \frac{\partial}{\partial y}(-x^2y^2) + \frac{\partial}{\partial z}(-x^2yz)$$

$$\nabla\cdot\vec{P} = 3x^2y - 2x^2y - x^2y = 0$$

Hence, $\vec{P}$ is solenoidal.
Now calculate curl:

$$\nabla\times\vec{P} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ x^3y & -x^2y^2 & -x^2yz \end{vmatrix}$$

$$\nabla\times\vec{P} = \hat{a}_x\left(\frac{\partial(-x^2yz)}{\partial y} - \frac{\partial(-x^2y^2)}{\partial z}\right) + \hat{a}_y\left(\frac{\partial(x^3y)}{\partial z} - \frac{\partial(-x^2yz)}{\partial x}\right) + \hat{a}_z\left(\frac{\partial(-x^2y^2)}{\partial x} - \frac{\partial(x^3y)}{\partial y}\right)$$

$$\nabla \times \vec{P} = (-x^2 z)\hat{a}_x + (2xyz)\hat{a}_y + (-2xy^2 - x^3)\hat{a}_z \neq 0$$

Therefore, $\vec{P}$ is not irrotational.

$$\boxed{\text{(a)}\vec{P} \text{ is solenoidal but not irrotational}}$$

Q4.14.1 NAT 2014 2M Ans: 3 ● ○ ○

Given,

$$\vec{r} = x\hat{a}_x + y\hat{a}_y + z\hat{a}_z$$

and

$$|\vec{r}| = r = \sqrt{x^2 + y^2 + z^2}$$

We know that

$$\nabla(\ln r) = \frac{1}{r}\nabla r$$

Also,

$$\nabla r = \frac{x\hat{a}_x + y\hat{a}_y + z\hat{a}_z}{r} = \frac{\vec{r}}{r}$$

Hence,

$$\nabla(\ln r) = \frac{\vec{r}}{r^2}$$

Therefore,

$$r^2\nabla(\ln r) = \vec{r}$$

Now,

$$\text{div}\left(r^2\nabla(\ln r)\right) = \nabla \cdot \vec{r}$$

$$= \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z}$$

$$= 1 + 1 + 1 = 3$$

$$\boxed{3}$$

Q4.14.2 NAT 2014 1M Ans: 3 ● ○ ○

Given,

$$f(x, y) = \frac{xy}{\sqrt{2}}(x + y)$$

The directional derivative in the direction of unit vector $\hat{u}$ is

$$D_{\hat{u}} f = \nabla f \cdot \hat{u}$$

First expand the function:

$$f(x, y) = \frac{x^2 y + xy^2}{\sqrt{2}}$$

Now calculate the gradient.

$$\frac{\partial f}{\partial x} = \frac{1}{\sqrt{2}}(2xy + y^2)$$

$$\frac{\partial f}{\partial y} = \frac{1}{\sqrt{2}}(x^2 + 2xy)$$

At (1, 1),

$$\nabla f(1,1) = \frac{1}{\sqrt{2}}(3\hat{a}_x + 3\hat{a}_y)$$

The unit vector makes an angle $\frac{\pi}{4}$ with the y-axis.
Hence,

$$\hat{u} = \sin\frac{\pi}{4}\,\hat{a}_x + \cos\frac{\pi}{4}\,\hat{a}_y = \frac{1}{\sqrt{2}}(\hat{a}_x + \hat{a}_y)$$

Therefore,

$$D_{\hat{u}} f = \nabla f \cdot \hat{u}$$

$$= \left(\frac{3}{\sqrt{2}}\hat{a}_x + \frac{3}{\sqrt{2}}\hat{a}_y\right)\cdot\left(\frac{1}{\sqrt{2}}\hat{a}_x + \frac{1}{\sqrt{2}}\hat{a}_y\right)$$

$$= \frac{3}{2} + \frac{3}{2} = 3$$

$$\boxed{3}$$

Q4.14.3 NAT 2014 2M Ans: 3.14 ● ○ ○

Given,

$$\vec{F} = z\hat{a}_x + x\hat{a}_y + y\hat{a}_z$$

We need to evaluate

$$\iint_S (\nabla \times \vec{F}) \cdot d\vec{S}$$

where S is the upper hemisphere

$$x^2 + y^2 + z^2 = 1, \qquad z \geq 0$$

Using Stokes' theorem,

$$\iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \oint_C \vec{F} \cdot d\vec{l}$$

where C is the boundary circle:

$$x^2 + y^2 = 1, \qquad z = 0$$

On the boundary,

$$\vec{F} = x\hat{a}_y + y\hat{a}_z$$

Parameterize the circle as

$$x = \cos\theta, \qquad y = \sin\theta, \qquad z = 0$$

Then,

$$d\vec{l} = (-\sin\theta\,\hat{a}_x + \cos\theta\,\hat{a}_y)\,d\theta$$

Hence,

$$\vec{F} \cdot d\vec{l} = (\cos\theta\,\hat{a}_y + \sin\theta\,\hat{a}_z) \cdot (-\sin\theta\,\hat{a}_x + \cos\theta\,\hat{a}_y)\,d\theta$$

$$= \cos^2\theta\,d\theta$$

Therefore,

$$\oint_C \vec{F} \cdot d\vec{l} = \int_0^{2\pi} \cos^2\theta\,d\theta$$

$$= \int_0^{2\pi} \frac{1 + \cos 2\theta}{2}\,d\theta = \pi$$

$$\boxed{3.14}$$

Q4.13.1 MCQ 2013 1M Ans: D ● ○ ○

Using Stokes' theorem, the closed loop line integral of a vector field $\vec{A}(\vec{r})$ is related to the surface integral of its curl over an open surface bounded by the loop

$$\oint \vec{A} \cdot d\vec{l} = \iint (\nabla \times \vec{A}) \cdot d\vec{s}$$

Here, the surface must be an **open surface** whose boundary is the given closed loop.
Hence, the correct expression is:

$$\oint \vec{A} \cdot d\vec{l} = \iint (\nabla \times \vec{A}) \cdot d\vec{s} \quad \text{over the open surface bounded by the loop which is given by option D.}$$

$$\boxed{\text{(d) } \oint \vec{A} \cdot d\vec{l} = \iint (\nabla \times \vec{A}) \cdot d\vec{s} \quad \text{over the open surface bounded by the loop}}$$

Q4.13.2 MCQ 2013 1M Ans: D ● ○ ○

Given the vector field

$$\vec{A} = x\hat{a}_x + y\hat{a}_y + z\hat{a}_z$$

The divergence of a vector field is defined as

$$\nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

Here,

$$A_x = x, \qquad A_y = y, \qquad A_z = z$$

Therefore,

$$\nabla \cdot \vec{A} = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z}$$

$$\nabla \cdot \vec{A} = 1 + 1 + 1 = 3$$

$$\boxed{\text{(d) } 3}$$

Q4.12.1 MCQ 2012 2M Ans: A ● ○ ○

Given that the vector $\vec{A}$ is radially outward,

$$|\vec{A}| = Kr^n$$

where

$$r = \sqrt{x^2 + y^2 + z^2}$$

Since the direction is radial,

$$\vec{A} = Kr^n \hat{a}_r$$

For a radial vector field,

$$\nabla \cdot \vec{A} = \frac{1}{r^2}\frac{d}{dr}(r^2 A_r)$$

Here,

$$A_r = Kr^n$$

Therefore,

$$\nabla \cdot \vec{A} = \frac{1}{r^2}\frac{d}{dr}(Kr^{n+2})$$

$$= \frac{1}{r^2}K(n+2)r^{n+1}$$

$$= K(n+2)r^{n-1}$$

Given,

$$\nabla \cdot \vec{A} = 0$$

Hence,

$$n + 2 = 0$$

$$n = -2$$

$$\boxed{\text{(a)} \ -2}$$

Q4.11.1 MCQ 2011 2M Ans: D ● ○ ○

Given,

$$\iint_S 5\vec{r} \cdot \hat{n}\, ds$$

Using Divergence theorem,

$$\iint_S \vec{A} \cdot \hat{n}\, ds = \iiint_V (\nabla \cdot \vec{A})\, dv$$

Here,

$$\vec{A} = 5\vec{r} = 5(x\hat{a}_x + y\hat{a}_y + z\hat{a}_z)$$

Therefore,

$$\nabla \cdot \vec{A} = \frac{\partial(5x)}{\partial x} + \frac{\partial(5y)}{\partial y} + \frac{\partial(5z)}{\partial z}$$

$$= 5 + 5 + 5 = 15$$

Hence,

$$\iint_S 5\vec{r} \cdot \hat{n}\, ds = \iiint_V 15\, dv$$

$$= 15V$$

$$\boxed{\text{(d) } 15V}$$

Q4.10.1 MCQ 2010 2M Ans: C ● ○ ○

Given,

$$\vec{A} = xy\,\hat{a}_x + x^2\,\hat{a}_y$$

The closed line integral is evaluated using Green's theorem:

$$\oint_C \vec{A} \cdot d\vec{l} = \oint_C (A_x\,dx + A_y\,dy) = \iint_S \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right) dx\,dy$$

Here, $A_x = xy, A_y = x^2$

Therefore,

$$\frac{\partial A_y}{\partial x} = 2x, \qquad \frac{\partial A_x}{\partial y} = x$$

Hence,

$$\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} = 2x - x = x$$

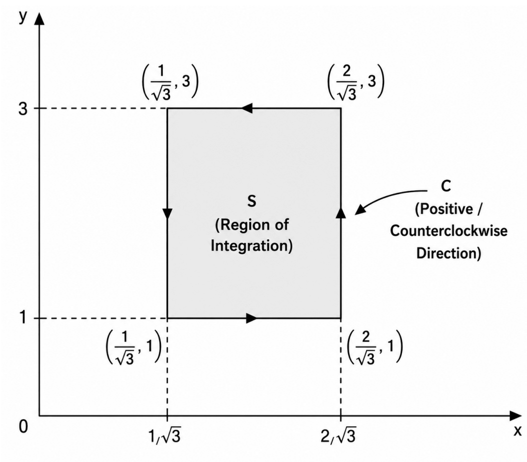

Fig. Solution Q4.10.1

The rectangular region is bounded by

$$\frac{1}{\sqrt{3}} \le x \le \frac{2}{\sqrt{3}}, \qquad 1 \le y \le 3$$

Thus,

$$\oint_C \vec{A} \cdot d\vec{l} = \int_{1/\sqrt{3}}^{2/\sqrt{3}} \int_1^3 x \, dy \, dx$$

$$= \int_{1/\sqrt{3}}^{2/\sqrt{3}} 2x \, dx$$

$$= \left[x^2\right]_{1/\sqrt{3}}^{2/\sqrt{3}} = \frac{4}{3} - \frac{1}{3} = 1$$

$\boxed{\text{(c) } 1}$

Q4.08.1 | MCQ | 2008 | 2M | Ans: B | ● ○ ○

Given,

$$P = (1, 0), \qquad Q = (0, 1)$$

We need to evaluate

$$2 \int_P^Q (x \, dx + y \, dy)$$

along the semicircular path having PQ as diameter.

Method 1

Observe that

$$x\,dx + y\,dy = \frac{1}{2}d(x^2 + y^2)$$

Therefore,

$$2\int_P^Q (x\,dx + y\,dy) = \int_P^Q d(x^2 + y^2)$$

Using the fundamental theorem for line integrals,

$$= \left(x^2 + y^2\right)_Q - \left(x^2 + y^2\right)_P$$

At point $Q = (0, 1)$,

$$x^2 + y^2 = 0^2 + 1^2 = 1$$

At point $P = (1, 0)$,

$$x^2 + y^2 = 1^2 + 0^2 = 1$$

Hence,

$$2\int_P^Q (x\,dx + y\,dy) = 1 - 1 = 0$$

Method 2

Since the variables of integration match the terms, integrate separately:

$$2\int_P^Q (x\,dx + y\,dy) = 2\int_P^Q x\,dx + 2\int_P^Q y\,dy$$

Now,

$$2\int x\,dx = x^2$$

and

$$2\int y\,dy = y^2$$

Therefore,

$$2\int_P^Q (x\,dx + y\,dy) = [x^2 + y^2]_P^Q$$

At $Q = (0, 1)$,

$$x^2 + y^2 = 0^2 + 1^2 = 1$$

At $P = (1, 0)$,

$$x^2 + y^2 = 1^2 + 0^2 = 1$$

Hence,

$$2\int_P^Q (x\,dx + y\,dy) = 1 - 1 = 0$$

$$\boxed{\text{(b) } 0}$$

Key Points

- The integrand is an exact differential:

$$d(x^2 + y^2) = 2x\,dx + 2y\,dy$$

- Line integrals of conservative vector fields are path independent.
- Only the initial and final points are needed for evaluation.

Q4.08.2 MCQ 2008 2M Ans: A ● ○ ○

Given,

$$g(x, y) = 4x^3 + 10y^4$$

The straight line joining $(0, 0)$ and $(1, 2)$ is $y = 2x$
Substituting this into the function,

$$g(x, 2x) = 4x^3 + 10(2x)^4$$

$$= 4x^3 + 160x^4$$

Now integrate from $x = 0$ to $x = 1$:

$$\int_0^1 \left(4x^3 + 160x^4\right)\,dx$$

$$= \left[x^4 + 32x^5\right]_0^1$$

$$= 1 + 32 = 33$$

$$\boxed{\text{(a) } 33}$$

Q4.06.1 MCQ 2006 1M Ans: A ● ○ ○

Using Stokes' theorem,

$$\iint (\nabla \times \vec{P}) \cdot d\vec{s} = \oint \vec{P} \cdot d\vec{l}$$

Thus, the given surface integral of curl is converted into a closed line integral of the vector field $\vec{P}$ around the boundary curve.

$$\boxed{\text{(a) } \oint P \bullet d\vec{l}}$$

Q4.93.1 NAT | 1993 | 2M | Ans: 1 | ● ○ ○

Given,

$$\vec{V} = x\cos^2 y\,\hat{i} + x^2 e^z\,\hat{j} + z\sin^2 y\,\hat{k}$$

The surface S is the closed surface of a unit cube bounded by

$$0 \le x \le 1, \qquad 0 \le y \le 1, \qquad 0 \le z \le 1$$

Using the divergence theorem,

$$\iint_S \vec{V} \cdot \hat{n}\, dS = \iiint_V (\nabla \cdot \vec{V})\, dV$$

First compute the divergence:

$$\nabla \cdot \vec{V} = \frac{\partial}{\partial x}(x\cos^2 y) + \frac{\partial}{\partial y}(x^2 e^z) + \frac{\partial}{\partial z}(z\sin^2 y)$$

$$= \cos^2 y + 0 + \sin^2 y$$

Using the identity,

$$\cos^2 y + \sin^2 y = 1$$

Hence,

$$\nabla \cdot \vec{V} = 1$$

Therefore,

$$\iint_S \vec{V} \cdot \hat{n}\, dS = \iiint_V 1\, dV$$

Since the volume of the unit cube is 1,

$$\iint_S \vec{V} \cdot \hat{n}\, dS = 1$$

$$\boxed{1}$$

Q4.26.1 NAT 2026 1M Ans: 0 ● ○ ○

Given,

$$\vec{F}(x, y, z) = \sin(y)\,\hat{x} + \cos(x)\,\hat{y} + 5\,\hat{z}$$

We need to evaluate

$$\oiint_S \vec{F}(x, y, z) \cdot d\vec{s}$$

where S is the closed unit sphere centered at the origin.
Using the divergence theorem,

$$\oiint_S \vec{F} \cdot d\vec{s} = \iiint_V (\nabla \cdot \vec{F})\, dV$$

Compute the divergence:

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(\sin y) + \frac{\partial}{\partial y}(\cos x) + \frac{\partial}{\partial z}(5)$$

$$= 0 + 0 + 0 = 0$$

Hence,

$$\oiint_S \vec{F} \cdot d\vec{s} = \iiint_V 0\, dV = 0$$

$$\boxed{0}$$

Mistakes to be avoided

- Do not differentiate $\sin y$ with respect to y while computing divergence; it must be differentiated with respect to x.
- Ensure the surface is closed before applying divergence theorem.
- The flux is not equal to the magnitude of the vector field.

Q4.25.1 NAT 2025 2M Ans: 0 ● ○ ○

Given,

$$V(x, y) = x^2 + x + y^2 + 1$$

The rate of change of a scalar function in a specified direction is given by the directional derivative:

$$D_{\hat{u}}V = \nabla V \cdot \hat{u}$$

First compute the gradient of V:

$$\nabla V = \frac{\partial V}{\partial x}\,\hat{i} + \frac{\partial V}{\partial y}\,\hat{j}$$

$$= (2x + 1)\,\hat{i} + (2y)\,\hat{j}$$

At the origin $(0, 0)$,

$$\nabla V(0, 0) = \hat{i}$$

The direction toward the point $(1, 2)$ is

$$\vec{a} = \hat{i} + 2\hat{j}$$

Its unit vector is

$$\hat{u} = \frac{\hat{i} + 2\hat{j}}{\sqrt{1^2 + 2^2}} = \frac{\hat{i} + 2\hat{j}}{\sqrt{5}}$$

Therefore,

$$D_{\hat{u}}V = \nabla V(0,0) \cdot \hat{u}$$

$$= \hat{i} \cdot \frac{\hat{i} + 2\hat{j}}{\sqrt{5}} = \frac{1}{\sqrt{5}}$$

$$\frac{1}{\sqrt{5}} \approx 0.447$$

Rounding to the nearest integer, answer becomes

$$\boxed{0}$$

Key Points

- The directional derivative measures the rate of change of a scalar field along a specified direction.
- The gradient vector points in the direction of maximum increase.
- The direction vector must always be converted into a unit vector before computing the directional derivative.

Q4.24.1 MCQ 2024 2M Ans: C ● ○ ○

Given,

$$\vec{u} = 2\hat{x} + \hat{y} + 2\hat{z}$$

and

$$f(x, y, z) = 2\ln(xy) + \ln(yz) + 3\ln(xz)$$

The directional derivative of f in the direction of $\vec{u}$ is

$$\nabla f \cdot \hat{u}$$

First, simplify the function:

$$f(x, y, z) = 2(\ln x + \ln y) + (\ln y + \ln z) + 3(\ln x + \ln z)$$

$$f(x, y, z) = 5\ln x + 3\ln y + 4\ln z$$

Hence,

$$\nabla f = \left(\frac{5}{x}, \frac{3}{y}, \frac{4}{z}\right)$$

At $(1, 1, 1)$,

$$\nabla f(1, 1, 1) = (5, 3, 4)$$

Now,

$$|\vec{u}| = \sqrt{2^2 + 1^2 + 2^2} = \sqrt{9} = 3$$

Therefore, the unit vector along $\vec{u}$ is

$$\hat{u} = \left(\frac{2}{3}, \frac{1}{3}, \frac{2}{3}\right)$$

Directional derivative:

$$\nabla f \cdot \hat{u} = (5, 3, 4) \cdot \left(\frac{2}{3}, \frac{1}{3}, \frac{2}{3}\right)$$

$$= \frac{10}{3} + \frac{3}{3} + \frac{8}{3} = \frac{21}{3} = 7$$

$$\boxed{\text{(c) } 7}$$

Q4.23.1 NAT 2023 2M Ans: 10 ● ○ ○

Given,

$$f(x_1, x_2) = x_1^2 + 2x_2^2 + 3x_1 + 3x_2 + x_1x_2 + 1$$

The magnitude of the maximum rate of change of a scalar function is equal to the magnitude of its gradient vector which is $|\nabla f|$

Compute the partial derivatives:

$$\frac{\partial f}{\partial x_1} = 2x_1 + x_2 + 3$$

$$\frac{\partial f}{\partial x_2} = 4x_2 + x_1 + 3$$

Hence,

$$\nabla f = (2x_1 + x_2 + 3,\ 4x_2 + x_1 + 3)$$

At the point $(1, 1)$,

$$\nabla f(1, 1) = (2 + 1 + 3,\ 4 + 1 + 3) = (6, 8)$$

Therefore,

$$|\nabla f(1, 1)| = \sqrt{6^2 + 8^2}$$

$$= \sqrt{36 + 64} = \sqrt{100} = 10$$

$$\boxed{10}$$

Q4.23.2 NAT 2023 2M Ans: 9.42 ● ● ○

Given vector field,

$$\mathbf{F} = -y\,\hat{i} + x\,\hat{j}$$

and the closed curve

$$r = 1 + \cos\theta$$

where

$$x = r\cos\theta, \qquad y = r\sin\theta$$

The required line integral is

$$\oint_C \mathbf{F} \cdot d\mathbf{r}$$

Using Green's theorem,

$$\oint_C (P\,dx + Q\,dy) = \iint_A \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$$

Here,

$$P = -y, \qquad Q = x$$

Therefore,

$$\frac{\partial Q}{\partial x} = 1, \qquad \frac{\partial P}{\partial y} = -1$$

Hence,

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1 - (-1) = 2$$

Thus,

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = 2 \times (\text{Area enclosed})$$

Area enclosed by the polar curve is

$$A = \frac{1}{2} \int_0^{2\pi} r^2 \, d\theta$$

Substituting

$$r = 1 + \cos\theta$$

$$A = \frac{1}{2} \int_0^{2\pi} (1 + \cos\theta)^2 \, d\theta$$

$$= \frac{1}{2} \int_0^{2\pi} \left(1 + 2\cos\theta + \cos^2\theta\right) d\theta$$

Using

$$\int_0^{2\pi} \cos\theta \, d\theta = 0, \qquad \int_0^{2\pi} \cos^2\theta \, d\theta = \pi$$

we get

$$A = \frac{1}{2}(2\pi + \pi) = \frac{3\pi}{2}$$

Therefore,

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = 2 \times \frac{3\pi}{2} = 3\pi$$

$$3\pi \approx 9.42$$

$$\boxed{9.42}$$

Key Points

- For a polar curve,

$$A = \frac{1}{2} \int r^2 \, d\theta$$

gives the enclosed area.

Q4.23.3 MCQ 2023 1M Ans: D ● ○ ○

For a hyperplane defined by

$$W^T X = 1$$

the vector normal to the plane is the coefficient vector itself.
Given,

$$W = [1, 2, 3]^T$$

Hence, the normal vector is

$$\vec{n} = [1, 2, 3]^T$$

$$\boxed{\text{(d) } [1, 2, 3]^T}$$

Key Points

- For a plane or hyperplane represented as $W^T X = c$, the vector W is always normal to the plane.
- The constant term on the RHS does not affect the direction of the normal vector.

Q4.22.1 MCQ 2022 2M Ans: C ● ● ○

Let the closed curve C enclosing the region R be traversed in the counterclockwise direction. Using Green's theorem, the area enclosed by a simple closed curve can be represented as

$$\text{Area}(R) = \oint_C x\,dy$$

Equivalent forms are

$$\text{Area}(R) = -\oint_C y\,dx$$

and

$$\text{Area}(R) = \frac{1}{2}\oint_C (x\,dy - y\,dx)$$

From Green's theorem,

$$\oint_C (P\,dx + Q\,dy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$$

Choosing

$$P = 0, \qquad Q = x$$

gives

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1$$

Hence,

$$\oint_C x\,dy = \iint_R 1\,dA = \text{Area}(R)$$

Choosing

$$P = -y, \qquad Q = 0$$

we obtain

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0 - (-1) = 1$$

Thus,

$$-\oint_C y\,dx = \text{Area}(R)$$

Combining both forms,

$$\text{Area}(R) = \frac{1}{2}\oint_C (x\,dy - y\,dx)$$

$$\boxed{\text{Checking each option, the incorrect option is 'C'}}$$

Key Points

- Area enclosed by a positively oriented closed curve can be evaluated using line integrals.
- Standard area formulas:

$$A = \oint_C x\,dy$$

$$A = -\oint_C y\,dx$$

$$A = \frac{1}{2}\oint_C (x\,dy - y\,dx)$$

- Counterclockwise orientation corresponds to positive area.

Mistakes to be avoided

- Do not ignore the orientation of the curve.
- Clockwise traversal changes the sign of the line integral.

Q4.22.2 MCQ 2022 2M Ans: C ● ○ ○

Given,

$$\vec{E}(x, y, z) = 2x^2\hat{i} + 5y\hat{j} + 3z\hat{k}$$

We need to evaluate

$$\iiint_V (\nabla \cdot \vec{E})\, dV$$

where V is the unit cube:

$$0 \le x \le 1, \qquad 0 \le y \le 1, \qquad 0 \le z \le 1$$

Compute the divergence:

$$\nabla \cdot \vec{E} = \frac{\partial}{\partial x}(2x^2) + \frac{\partial}{\partial y}(5y) + \frac{\partial}{\partial z}(3z)$$

$$= 4x + 5 + 3 = 4x + 8$$

Hence,

$$\iiint_V (4x + 8)\, dV$$

Using the limits of the unit cube,

$$\int_0^1\int_0^1\int_0^1 (4x + 8)\, dx\, dy\, dz$$

First integrate with respect to x:

$$\int_0^1 (4x + 8)\, dx = \left[2x^2 + 8x\right]_0^1 = 2 + 8 = 10$$

Now,

$$\int_0^1\int_0^1 10\, dy\, dz = 10$$

$$\boxed{10}$$

Q4.22.3 MCQ 2022 2M Ans: B ● ○ ○

Given,

$$f(x, y, z) = 4x^2 + 7xy + 3xz^2$$

The function increases most rapidly in the direction of its gradient vector given by ∇f
Compute the partial derivatives:

$$\frac{\partial f}{\partial x} = 8x + 7y + 3z^2, \frac{\partial f}{\partial y} = 7x, \frac{\partial f}{\partial z} = 6xz$$

,

Hence,

$$\nabla f = \left(8x + 7y + 3z^2\right)\hat{i} + 7x\hat{j} + 6xz\hat{k}$$

At the point

$$P = (1, 0, 2)$$

we get

$$\nabla f(1, 0, 2) = (8 + 0 + 12)\hat{i} + 7\hat{j} + 12\hat{k}$$

$$= 20\hat{i} + 7\hat{j} + 12\hat{k}$$

Therefore, the direction of maximum increase is

$$20\hat{i} + 7\hat{j} + 12\hat{k}$$

$$\boxed{\text{(b) } 20\hat{i} + 7\hat{j} + 12\hat{k}}$$

Q4.20.1 NAT 2020 2M Ans: -3 ● ○ ○

Given vector field,

$$\vec{F} = a_x y - a_y x$$

The line integral is

$$\int_C \vec{F} \cdot d\vec{l}$$

Along the parabola,

$$y = x^2$$

From the figure, the path is from $(-1, 1)$ to $(2, 4)$.
Now,

$$\vec{F} \cdot d\vec{l} = y\,dx - x\,dy$$

Using

$$y = x^2, \qquad dy = 2x\,dx$$

we get

$$\vec{F} \cdot d\vec{l} = x^2 dx - x(2x\,dx)$$

$$\vec{F} \cdot d\vec{l} = -x^2 dx$$

Hence,

$$\int_C \vec{F} \cdot d\vec{l} = \int_{-1}^{2} (-x^2)\,dx$$

$$= -\left[\frac{x^3}{3}\right]_{-1}^{2}$$

$$= -\left(\frac{8}{3} + \frac{1}{3}\right)$$

$$= -3$$

$$\boxed{-3.00}$$

Q4.20.2 MCQ 2020 2M Ans: B ● ○ ○

Given vector field,

$$\vec{F} = a_x(5y - k_1 z) + a_y(3z + k_2 x) + a_z(k_3 y - 4x)$$

For a conservative field,

$$\nabla \times \vec{F} = 0$$

Using curl condition,

$$\frac{\partial F_z}{\partial y} = \frac{\partial F_y}{\partial z}$$

$$k_3 = 3$$

Also,

$$\frac{\partial F_x}{\partial z} = \frac{\partial F_z}{\partial x}$$

$$-k_1 = -4$$

$$k_1 = 4$$

Further,

$$\frac{\partial F_y}{\partial x} = \frac{\partial F_x}{\partial y}$$

$$k_2 = 5$$

Hence,

$$k_1 = 4, \qquad k_2 = 5, \qquad k_3 = 3$$

$$\boxed{\text{(b) } k_1 = 4,\ k_2 = 5,\ k_3 = 3}$$

Q4.19.1 NAT 2019 1M Ans: 139 ● ● ○

Given,

$$f = 2x^3 + 3y^2 + 4z$$

We need to evaluate

$$\int_C \nabla f \cdot d\mathbf{r}$$

where the contour C joins the points

$$(-3, -3, 2) \to (2, -3, 2) \to (2, 6, 2) \to (2, 6, -1)$$

Since the vector field is a gradient field,

$$\int_C \nabla f \cdot d\mathbf{r} = f(B) - f(A)$$

where

$$A = (-3, -3, 2), \qquad B = (2, 6, -1)$$

Evaluate f at the endpoints.
At $A = (-3, -3, 2)$:

$$f(A) = 2(-3)^3 + 3(-3)^2 + 4(2)$$

$$= 2(-27) + 3(9) + 8$$

$$= -54 + 27 + 8 = -19$$

At $B = (2, 6, -1)$:

$$f(B) = 2(2)^3 + 3(6)^2 + 4(-1)$$

$$= 16 + 108 - 4$$

$$= 120$$

Therefore,

$$\int_C \nabla f \cdot d\mathbf{r} = 120 - (-19) = 139$$

$$\boxed{139}$$

Key Points

- For a scalar function f,

$$\int_C \nabla f \cdot d\mathbf{r} = f(\text{final point}) - f(\text{initial point})$$

Q4.18.1 NAT 2018 2M Ans: 0 ● ● ○

Method 1

Given,

$$\int_C \left[(y^2 + 2yx)\,dx + (2xy + x^2)\,dy\right]$$

where C is the arc of the circle $x^2 + y^2 = 9$ from $(3, 0)$ to $(0, 3)$.
Let

$$M = y^2 + 2yx, \qquad N = 2xy + x^2$$

Check whether the differential form is exact or not: $\frac{\partial M}{\partial y} = 2y + 2x$ and $\frac{\partial N}{\partial x} = 2y + 2x$
Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ the integral is path independent.
Hence, there exists a scalar function $\phi(x, y)$ such that

$$d\phi = M\,dx + N\,dy$$

Integrate M with respect to x:

$$\phi(x, y) = \int (y^2 + 2yx)\,dx$$

$$= xy^2 + x^2y + g(y)$$

Differentiate with respect to y:

$$\frac{\partial \phi}{\partial y} = 2xy + x^2 + g'(y)$$

Comparing with

$$N = 2xy + x^2$$

gives

$$g'(y) = 0$$

Therefore,

$$\phi(x, y) = xy^2 + x^2y$$

Now evaluate at the endpoints.
At $(0, 3)$:

$$\phi(0, 3) = 0$$

At $(3, 0)$:

$$\phi(3, 0) = 0$$

Thus,

$$\int_C \left[(y^2 + 2yx)\,dx + (2xy + x^2)\,dy\right] = 0 - 0 = 0$$

Method 2

Parameterize the circular arc using

$$x = r\cos\theta, \qquad y = r\sin\theta$$

Since $x^2 + y^2 = 9$ we have $r = 3$
Hence,

$$x = 3\cos\theta, \qquad y = 3\sin\theta$$

The curve starts at $(3, 0) \Rightarrow \theta = 0$ and ends at $(0, 3) \Rightarrow \theta = \frac{\pi}{2}$

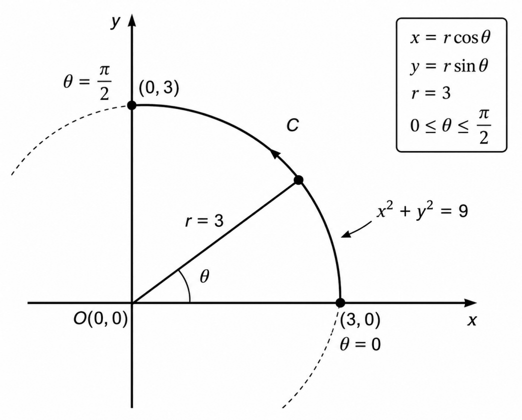

Fig. Solution Q4.18.1

Now, $dx = -3\sin\theta\,d\theta$, $dy = 3\cos\theta\,d\theta$
Substitute into the integral, we get

$$y^2 + 2yx = (3\sin\theta)^2 + 2(3\sin\theta)(3\cos\theta) = 9\sin^2\theta + 18\sin\theta\cos\theta$$

Also,

$$2xy + x^2 = 2(3\cos\theta)(3\sin\theta) + (3\cos\theta)^2 = 18\sin\theta\cos\theta + 9\cos^2\theta$$

Therefore,

$$I = \int_0^{\pi/2} \big[(9\sin^2\theta + 18\sin\theta\cos\theta)(-3\sin\theta)$$
$$+(18\sin\theta\cos\theta + 9\cos^2\theta)(3\cos\theta)\big]\,d\theta$$

$$I = 27\int_0^{\pi/2} \big[-\sin^3\theta - 2\sin^2\theta\cos\theta + 2\sin\theta\cos^2\theta + \cos^3\theta\big]\,d\theta$$

Observe that $\frac{d}{d\theta}(\sin\theta\cos\theta) = \cos^2\theta - \sin^2\theta$
and the integrand simplifies to an exact derivative:

$$I = 27\int_0^{\pi/2} \frac{d}{d\theta}(\sin\theta\cos\theta)\,d\theta$$

Hence,

$$I = 27\,[\sin\theta\cos\theta]_0^{\pi/2}$$

$$= 27(0-0) = 0$$

$$\boxed{0}$$

Key Points

- A line integral is path independent if
$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$
in a simply connected region.
- For an exact differential,
$$\int_C (M\,dx + N\,dy) = \phi(B) - \phi(A)$$
- Only the endpoints are needed once the potential function is found.

Q4.18.2 MCQ 2018 1M Ans: A ● ● ○

Given,

$$\phi(x,y,z) = xy^2 + yz^2 + zx^2$$

Point:

$$P = (2,-1,1)$$

Direction vector:

$$\mathbf{p} = \hat{\imath} + 2\hat{\jmath} + 2\hat{k}$$

The directional derivative in the direction of **p** is

$$D_{\mathbf{p}}\phi = \nabla\phi\cdot\hat{u}$$

where $\hat{u}$ is the unit vector along **p**.
Compute the gradient:

$$\frac{\partial\phi}{\partial x} = y^2 + 2zx,\ \frac{\partial\phi}{\partial y} = 2xy + z^2,\ \frac{\partial\phi}{\partial z} = 2yz + x^2$$

Hence,

$$\nabla\phi = (y^2+2zx)\hat{\imath} + (2xy+z^2)\hat{\jmath} + (2yz+x^2)\hat{k}$$

At the point $(2,-1,1)$:

$$\nabla\phi(2,-1,1) = \big[(-1)^2 + 2(1)(2)\big]\,\hat{\imath} + \big[2(2)(-1) + 1^2\big]\,\hat{\jmath} + \big[2(-1)(1) + 2^2\big]\,\hat{k} = 5\hat{\imath} - 3\hat{\jmath} + 2\hat{k}$$

Now compute the unit vector along **p**:

$$|\mathbf{p}| = \sqrt{1^2 + 2^2 + 2^2} = 3$$

$$\hat{u} = \frac{1}{3}(\hat{\imath} + 2\hat{\jmath} + 2\hat{k})$$

Therefore,

$$D_{\mathbf{p}}\phi = (5\hat{\imath} - 3\hat{\jmath} + 2\hat{k}) \cdot \frac{1}{3}(\hat{\imath} + 2\hat{\jmath} + 2\hat{k})$$

$$= \frac{1}{3}(5 - 6 + 4) = \frac{3}{3} = 1$$

$$\boxed{\text{(a) } 1}$$

Q4.16.1 | NAT | 2016 | 2M | Ans: 4.42 | ● ○ ○

The vector field is

$$\vec{F} = 5xz\,\hat{\imath} + (3x^2 + 2y)\,\hat{\jmath} + x^2z\,\hat{k}$$

The path is parameterized by

$$x = t, \qquad y = t^2, \qquad z = t$$

with $0 \le t \le 1$.
Hence,

$$d\vec{r} = \left(\frac{dx}{dt}\hat{\imath} + \frac{dy}{dt}\hat{\jmath} + \frac{dz}{dt}\hat{k}\right) dt = (\hat{\imath} + 2t\,\hat{\jmath} + \hat{k})dt$$

Substituting $x = t,\ y = t^2,\ z = t$ into $\vec{F}$,

$$\vec{F} = 5t^2\,\hat{\imath} + 5t^2\,\hat{\jmath} + t^3\,\hat{k}$$

Now,

$$\vec{F} \cdot d\vec{r} = (5t^2)(1) + (5t^2)(2t) + (t^3)(1)$$

$$\vec{F} \cdot d\vec{r} = 5t^2 + 11t^3$$

Therefore, the line integral is

$$\int_0^1 (5t^2 + 11t^3)\,dt = \left[\frac{5t^3}{3} + \frac{11t^4}{4}\right]_0^1$$

$$= \frac{5}{3} + \frac{11}{4} = \frac{53}{12} = 4.42$$

$$\boxed{4.42}$$

Q4.16.2 | MCQ | 2016 | 1M | Ans: B | ● ○ ○

Given line integral is

$$\int_C \left(2xy^2\,dx + 2x^2y\,dy + dz\right)$$

Observe that

$$2xy^2\,dx + 2x^2y\,dy + dz = d(x^2y^2 + z)$$

Hence, the integral is path independent and

$$\int_C d(x^2y^2 + z) = \left[x^2y^2 + z\right]_{(0,0,0)}^{(1,1,1)}$$

$$= (1^2 \cdot 1^2 + 1) - (0 + 0)$$

$$= 2$$

$$\boxed{\text{(b) } 2}$$

Q4.15.1 NAT 2015 2M Ans: 0.718 ● ○ ○

The required volume is

$$V = \iint_R e^x \, dA$$

The region is bounded by

$$x = y, \qquad x = 0, \qquad y = 1$$

Hence, the limits are

$$0 \le y \le 1, \qquad 0 \le x \le y$$

Therefore,

$$V = \int_0^1 \int_0^y e^x \, dx \, dy$$

Integrating with respect to x,

$$V = \int_0^1 [e^x]_0^y \, dy$$

$$= \int_0^1 (e^y - 1) \, dy$$

$$= [e^y - y]_0^1$$

$$= (e - 1) - 1$$

$$= e - 2 = 0.71$$

$$\boxed{0.71}$$

Q4.14.1 MCQ 2014 1M Ans: A ● ● ○

Given,

$$\nabla \cdot (f\vec{v}) = x^2 y + y^2 z + z^2 x$$

and

$$\vec{v} = y\hat{i} + z\hat{j} + x\hat{k}$$

Using the vector identity,

$$\nabla \cdot (f\vec{v}) = f(\nabla \cdot \vec{v}) + \vec{v} \cdot \nabla f$$

Now,

$$\nabla \cdot \vec{v} = \frac{\partial y}{\partial x} + \frac{\partial z}{\partial y} + \frac{\partial x}{\partial z} = 0$$

Hence,

$$\vec{v} \cdot \nabla f = \nabla \cdot (f\vec{v})$$

Therefore,

$$\vec{v} \cdot \nabla f = x^2 y + y^2 z + z^2 x$$

$$\boxed{\text{(a) } \vec{v} \cdot \nabla f = x^2 y + y^2 z + z^2 x}$$

Key Points

- Use the identity:
$$\nabla \cdot (f\vec{v}) = f(\nabla \cdot \vec{v}) + \vec{v} \cdot \nabla f$$
- If $\nabla \cdot \vec{v} = 0$, then
$$\vec{v} \cdot \nabla f = \nabla \cdot (f\vec{v})$$

Q4.14.2 MCQ 2014 2M Ans: B ● ○ ○

Given vector field,

$$\vec{F} = yz\,\hat{i}$$

The curve is the circle

$$x^2 + y^2 = 1, \qquad z = 1$$

At $z = 1$,

$$\vec{F} = y\,\hat{i}$$

The line integral is

$$\oint \vec{F} \cdot d\vec{r}$$

Now,

$$d\vec{r} = dx\,\hat{i} + dy\,\hat{j}$$

Hence,

$$\vec{F} \cdot d\vec{r} = y\,dx$$

Parameterize the unit circle in counterclockwise direction:

$$x = \cos\theta, \qquad y = \sin\theta$$

Then,

$$dx = -\sin\theta\,d\theta$$

Therefore,

$$\oint y\,dx = \int_0^{2\pi} \sin\theta(-\sin\theta)\,d\theta$$

$$= -\int_0^{2\pi} \sin^2\theta\,d\theta = -\pi$$

$$\boxed{\text{(b) } -\pi}$$

Q4.13.1 MCQ 2013 1M Ans: B ● ○ ○

Given,

$$\vec{F} = y^2 x\,\hat{a}_x - yz\,\hat{a}_y - x^2\,\hat{a}_z$$

The path is along the x-axis from $x = 1$ to $x = 2$.
Hence, on the path,

$$y = 0, \qquad z = 0, \qquad d\vec{l} = dx\,\hat{a}_x$$

Substituting $y = 0$ and $z = 0$ into $\vec{F}$,

$$\vec{F} = 0\,\hat{a}_x + 0\,\hat{a}_y - x^2\,\hat{a}_z$$

Now,

$$\vec{F} \cdot d\vec{l} = (0\,\hat{a}_x - x^2\,\hat{a}_z) \cdot (dx\,\hat{a}_x) = 0$$

Therefore,

$$\int \vec{F} \cdot d\vec{l} = 0$$

$$\boxed{\text{(b) } 0}$$

Q4.13.2 MCQ 2013 1M Ans: D ● ○ ○

Given scalar field,

$$V = 2x^2 y + 3y^2 z + 4z^2 x$$

We know that

$$\nabla \times (\nabla V) = 0$$

for any scalar field V having continuous second order derivatives.
Hence,

$$\text{curl}(\nabla V) = 0$$

$$\boxed{\text{(d) } 0}$$

Q4.12.1 MCQ 2012 2M Ans: A ● ○ ○

The vector $\vec{A}$ is radially outward, so its direction is along $\hat{r}$.
Given,

$$|\vec{A}| = kr^n$$

Hence,

$$\vec{A} = kr^n\hat{r}$$

Using the divergence formula in spherical coordinates for a purely radial vector field:

$$\nabla \cdot \vec{A} = \frac{1}{r^2}\frac{d}{dr}\left(r^2 A_r\right)$$

Here,

$$A_r = kr^n$$

Therefore,

$$\nabla \cdot \vec{A} = \frac{1}{r^2}\frac{d}{dr}\left(kr^{n+2}\right)$$

$$= \frac{1}{r^2}\left(k(n+2)r^{n+1}\right)$$

$$= k(n+2)r^{n-1}$$

For
$$\nabla \cdot \vec{A} = 0$$
we must have
$$n + 2 = 0$$
Hence,
$$n = -2$$

$$\boxed{\text{(a)} -2}$$

Q4.11.1 MCQ 2011 1M Ans: B ● ○ ○

Given vectors
$$\vec{v}_1 = [1, 1, 1]$$
and
$$\vec{v}_2 = [1, a, a^2]$$
where
$$a = -\frac{1}{2} + j\frac{\sqrt{3}}{2}$$
Since a is a cube root of unity,
$$1 + a + a^2 = 0$$
Now compute the dot product:
$$\vec{v}_1 \cdot \vec{v}_2 = 1 + a + a^2 = 0$$
Hence, the two vectors are orthogonal.
To be orthonormal, both vectors must also have unit magnitude, which is not true here.

$$\boxed{\text{(b) Orthogonal}}$$

Key Points

- Cube roots of unity satisfy
$$1 + a + a^2 = 0$$
- If the dot product of two vectors is zero, the vectors are orthogonal.
- Orthonormal vectors must be both orthogonal and unit vectors.

Q4.10.1 MCQ 2010 1M Ans: A ● ○ ○

The three-dimensional radial vector is
$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$
Its divergence is
$$\nabla \cdot \vec{r} = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z}$$
$$= 1 + 1 + 1$$
$$= 3$$

$$\boxed{\text{(a) } 3}$$

Key Points

- For $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$,
$$\nabla \cdot \vec{r} = 3$$
- In n-dimensions, divergence of the position vector equals n.

Q4.09.1 | MCQ | 2009 | 2M | Ans: D | ● ○ ○

Given,
$$\vec{F}(x, y) = (x^2 + xy)\hat{a}_x + (y^2 + xy)\hat{a}_y$$
The straight line joining $(0, 2)$ and $(2, 0)$ is
$$x + y = 2$$
Hence,
$$y = 2 - x$$
and
$$dy = -dx$$
The line integral is
$$\int_C \vec{F} \cdot d\vec{r} = \int_C \left[(x^2 + xy)\, dx + (y^2 + xy)\, dy\right]$$
Substituting $y = 2 - x$,
$$x^2 + x(2 - x) = 2x$$
Also,
$$y^2 + xy = (2 - x)^2 + x(2 - x) = 4 - 2x$$
Therefore,
$$\int_0^2 [2x\, dx + (4 - 2x)(-dx)]$$
$$= \int_0^2 (4x - 4)\, dx$$
$$= \left[2x^2 - 4x\right]_0^2$$
$$= 8 - 8 = 0$$
$$\boxed{\text{(d) } 0}$$

Q4.06.1 | MCQ | 2006 | 2M | Ans: D | ● ○ ○

Given surface,
$$S(x, y) = 2x + 5y - 3$$
The path is
$$(x + 1)^2 + (y - 1)^2 = \sqrt{2}$$
which represents a closed curve.
The required integral is
$$\oint S(x, y) dS$$
Since S is a scalar function, its exact differential integrated over any closed path is zero.

Hence,

$$\oint S(x,y)dS = 0$$

$$\boxed{\text{(d) } 0}$$

Q4.05.1 MCQ 2005 2M Ans: C ● ○ ○

Given,

$$u = \frac{x^2}{2} + \frac{y^2}{3}$$

Gradient of u is

$$\nabla u = \frac{\partial u}{\partial x}\hat{i} + \frac{\partial u}{\partial y}\hat{j}$$

$$\nabla u = x\hat{i} + \frac{2y}{3}\hat{j}$$

At the point $(1, 3)$,

$$\nabla u = \hat{i} + 2\hat{j}$$

Magnitude of the gradient is

$$|\nabla u| = \sqrt{1^2 + 2^2}$$

$$= \sqrt{5}$$

$$\boxed{\text{(c) } \sqrt{5}}$$

Q4.25.1 MCQ 2025 2M Ans: B ● ○ ○

Using the divergence theorem,

$$\iint_S [(2x+z)\,dy\,dz + (2x+z)\,dx\,dz + (2z+y)\,dx\,dy] = \iiint_V \nabla\cdot\vec{F}\,dV$$

where

$$\vec{F} = (2x+z)\,\hat{\imath} + (2x+z)\,\hat{\jmath} + (2z+y)\,\hat{k}.$$

Compute the divergence:

$$\nabla\cdot\vec{F} = \frac{\partial}{\partial x}(2x+z) + \frac{\partial}{\partial y}(2x+z) + \frac{\partial}{\partial z}(2z+y)$$
$$= 2+0+2 = 4.$$

The region is the sphere

$$x^2+y^2+z^2 = 9$$

with radius

$$r = 3.$$

Hence,

$$\text{Volume of sphere} = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(3)^3 = 36\pi.$$

Therefore,

$$\iint_S \vec{F}\cdot d\vec{S} = \iiint_V 4\,dV = 4(36\pi) = 144\pi.$$

$$\boxed{\text{(b) } 144\pi}$$

Q4.20.1 MCQ 2020 1M Ans: C ● ○ ○

Since vectors $\vec{a}$ and $\vec{b}$ lie on the plane $z=0$, both vectors are confined to the xy-plane.
Also, given that

$$\vec{a} \neq \alpha\vec{b},$$

the vectors are non-parallel. Hence, their cross product is non-zero and perpendicular to the xy-plane.
A vector perpendicular to the xy-plane is along the z-axis, i.e. $\hat{k}$.

$$\boxed{\text{(c) } \hat{k}}$$

Q4.19.1 MCQ 2019 1M Ans: C ● ○ ○

Since $\vec{a},\vec{b},\vec{c}$ are mutually orthogonal vectors, the vector $\vec{c}$ must be perpendicular to both $\vec{a}$ and $\vec{b}$.
A vector perpendicular to both $\vec{a}$ and $\vec{b}$ is given by their cross product:

$$\vec{c} \parallel \vec{a}\times\vec{b}.$$

Given,

$$\vec{a} = \hat{\imath} + 2\hat{\jmath} + 5\hat{k}, \qquad \vec{b} = \hat{\imath} + 2\hat{\jmath} - \hat{k}.$$

Now,

$$\vec{a}\times\vec{b} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 1 & 2 & 5 \\ 1 & 2 & -1 \end{vmatrix}$$

$$= \hat{\imath}(2\cdot(-1) - 5\cdot 2) - \hat{\jmath}(1\cdot(-1) - 5\cdot 1) + \hat{k}(1\cdot 2 - 2\cdot 1)$$

$$= -12\hat{i} + 6\hat{j} + 0\hat{k} = -6(2\hat{i} - \hat{j}).$$

Hence,

$$\vec{c} \parallel (2\hat{i} - \hat{j}).$$

$$\boxed{\text{(c) } 2\hat{i} - \hat{j}}$$

Q4.18.1 | MCQ | 2018 | 2M | Ans: D | ● ○ ○

The vector field is

$$\vec{F} = (x^2 - 2y)\hat{i} - 4yz\hat{j} + 4xz^2\hat{k}.$$

The straight line from $(0, 0, 0)$ to $(1, 1, 1)$ can be parameterized as

$$x = t, \quad y = t, \quad z = t, \qquad 0 \le t \le 1.$$

Hence,

$$d\vec{l} = dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k} = (\hat{i} + \hat{j} + \hat{k})dt.$$

Substituting $x = y = z = t$ in $\vec{F}$:

$$\vec{F} = (t^2 - 2t)\hat{i} - 4t^2\hat{j} + 4t^3\hat{k}.$$

Now,

$$\vec{F} \cdot d\vec{l} = \left[(t^2 - 2t) - 4t^2 + 4t^3\right] dt$$

$$= (4t^3 - 3t^2 - 2t)\, dt.$$

Therefore,

$$\int_C \vec{F} \cdot d\vec{l} = \int_0^1 (4t^3 - 3t^2 - 2t)\, dt$$

$$= \left[t^4 - t^3 - t^2\right]_0^1 = 1 - 1 - 1 = -1.$$

$$\boxed{(d) - 1}$$

Q4.17.1 | NAT | 2017 | 2M | Ans: 0.72 | ● ○ ○

Let

$$X_1 = \begin{bmatrix} 2 \\ 6 \\ 14 \end{bmatrix}, \qquad X_2 = \begin{bmatrix} -12 \\ 8 \\ 16 \end{bmatrix}.$$

Using the formula for angle between two vectors,

$$\cos\theta = \frac{X_1 \cdot X_2}{|X_1||X_2|}.$$

Compute the dot product:

$$X_1 \cdot X_2 = (2)(-12) + (6)(8) + (14)(16)$$

$$= -24 + 48 + 224 = 248.$$

Now,

$$|X_1| = \sqrt{2^2 + 6^2 + 14^2} = \sqrt{236},$$

$$|X_2| = \sqrt{(-12)^2 + 8^2 + 16^2} = \sqrt{464}.$$

Hence,

$$\cos\theta = \frac{248}{\sqrt{236}\sqrt{464}} = \frac{248}{8\sqrt{1711}} = \frac{31}{\sqrt{1711}}.$$

Therefore,

$$\theta = \cos^{-1}\left(\frac{31}{\sqrt{1711}}\right) = 0.72$$

$$\boxed{0.72}$$

Q4.16.1 MCQ 2016 1M Ans: D ● ○ ○

Let

$$\vec{a} = \hat{\imath} + \hat{\jmath} + \hat{k}, \qquad \vec{b} = \hat{\imath} + 2\hat{\jmath} + 3\hat{k}.$$

A vector perpendicular to both $\vec{a}$ and $\vec{b}$ must satisfy

$$\vec{v}\cdot\vec{a} = 0 \quad \text{and} \quad \vec{v}\cdot\vec{b} = 0.$$

Check option D:

$$\vec{v} = 4\hat{\imath} + 3\hat{\jmath} + 5\hat{k}.$$

Now,

$$\vec{v}\cdot\vec{a} = 4 + 3 + 5 = 12 \neq 0.$$

Hence, this vector is not perpendicular to $\vec{a}$ and therefore not perpendicular to both vectors.
For verification:
Option A:

$$(1,-2,1)\cdot(1,1,1) = 1 - 2 + 1 = 0,$$
$$(1,-2,1)\cdot(1,2,3) = 1 - 4 + 3 = 0.$$

Option B:

$$(-1,2,-1)\cdot(1,1,1) = -1 + 2 - 1 = 0,$$
$$(-1,2,-1)\cdot(1,2,3) = -1 + 4 - 3 = 0.$$

Option C:

$$(0,0,0)$$

is perpendicular to every vector.

$$\boxed{\text{(d) } 4\hat{\imath} + 3\hat{\jmath} + 5\hat{k}}$$

Q4.15.1 MCQ 2015 1M Ans: A ● ○ ○

Given

$$f(x,y) = x^2 + 3y^2.$$

The directional derivative in the direction normal to the circle

$$x^2 + y^2 = 2$$

is along the gradient of

$$g(x,y) = x^2 + y^2.$$

First find the gradient of f:

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right) = (2x, 6y).$$

At the point $(1,1)$,

$$\nabla f(1,1) = (2,6).$$

Now,

$$\nabla g = (2x, 2y).$$

At $(1, 1)$,

$$\nabla g(1, 1) = (2, 2).$$

The unit normal vector is

$$\hat{n} = \frac{(2, 2)}{\sqrt{2^2 + 2^2}} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right).$$

Hence, the directional derivative is

$$D_{\hat{n}} f = \nabla f \cdot \hat{n}$$

$$= (2, 6) \cdot \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

$$= \frac{2 + 6}{\sqrt{2}} = \frac{8}{\sqrt{2}} = 4\sqrt{2}.$$

$$\boxed{\text{(a) } 4\sqrt{2}}$$

Q4.14.1 NAT | 2014 | 1M | Ans: 1 | ● ○ ○

Given vector field

$$\vec{f} = y\hat{i} + x\hat{j} + z\hat{k}.$$

Using Gauss divergence theorem,

$$\iint_S \vec{f} \cdot d\vec{s} = \iiint_V (\nabla \cdot \vec{f})\, dV.$$

Now,

$$\nabla \cdot \vec{f} = \frac{\partial y}{\partial x} + \frac{\partial x}{\partial y} + \frac{\partial z}{\partial z}$$

$$= 0 + 0 + 1 = 1.$$

The given surface is a cube with vertices from

$$(0, 0, 0) \text{ to } (1, 1, 1),$$

hence side length

$$= 1.$$

Therefore,

$$\text{Volume of cube} = 1^3 = 1.$$

Thus,

$$\iint_S \vec{f} \cdot d\vec{s} = \iiint_V 1\, dV = 1.$$

$$\boxed{1}$$

Q4.13.1 MCQ 2013 1M Ans: D ● ○ ○

For a vector field $\vec{E}$:

$$\nabla \cdot \vec{E} = 0$$

implies that the field is **solenoidal**.
Also,

$$\nabla \times \vec{E} = 0$$

implies that the field is **irrotational**. In a simply connected region, such a field is also conservative.
Check each option:

A. True, since zero divergence implies solenoidal field.

B. True, since zero curl implies conservative field.

C. True, since zero curl implies irrotational field.

D. False, because zero divergence does not imply irrotational field.

$$\boxed{\text{(d) If } \nabla \cdot E = 0, E \text{ is called irrotational}}$$

Q4.12.1 MCQ 2012 2M Ans: A ● ● ○

The vector $\vec{A}$ is directed radially outward, so it can be written as

$$\vec{A} = kr^n \hat{r}.$$

For a radial vector field of the form

$$\vec{F} = f(r)\hat{r},$$

the divergence in spherical coordinates is

$$\nabla \cdot \vec{F} = \frac{1}{r^2}\frac{d}{dr}\left(r^2 f(r)\right).$$

Here,

$$f(r) = kr^n.$$

Thus,

$$\nabla \cdot \vec{A} = \frac{1}{r^2}\frac{d}{dr}\left(r^2 \cdot kr^n\right)$$

$$= \frac{1}{r^2}\frac{d}{dr}\left(kr^{n+2}\right)$$

$$= \frac{1}{r^2}k(n+2)r^{n+1}$$

$$= k(n+2)r^{n-1}.$$

For

$$\nabla \cdot \vec{A} = 0,$$

we require

$$n + 2 = 0.$$

Hence,

$$n = -2.$$

$$\boxed{\text{(a) -2}}$$

Q4.09.1 MCQ 2009 1M Ans: A ● ○ ○

The sphere of unit radius centered at the origin is given by

$$x^2 + y^2 + z^2 = 1.$$

For a sphere centered at the origin, the outward normal vector at a point (x, y, z) is along the radius vector. The position vector of the point is

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}.$$

Its magnitude on the unit sphere is

$$|\vec{r}| = \sqrt{x^2 + y^2 + z^2} = 1.$$

Hence, the unit normal vector is

$$\hat{n} = \frac{\vec{r}}{|\vec{r}|} = (x, y, z).$$

$\boxed{\text{(a) (x,y,z)}}$

Q4.26.1 MCQ 2026 1M Ans: A ● ● ○

The region A is bounded by the parabola

$$x^2 = 4y \;\Rightarrow\; y = \frac{x^2}{4},$$

the line $x = 2$, and the x-axis.
For $0 \le x \le 2$, the limits of y are

$$0 \le y \le \frac{x^2}{4}.$$

Hence,

$$\iint_A y\,dx\,dy = \int_0^2 \int_0^{x^2/4} y\,dy\,dx.$$

Evaluating the inner integral,

$$\int_0^{x^2/4} y\,dy = \left[\frac{y^2}{2}\right]_0^{x^2/4} = \frac{x^4}{32}.$$

Therefore,

$$\int_0^2 \frac{x^4}{32}\,dx = \frac{1}{32}\left[\frac{x^5}{5}\right]_0^2 = \frac{32}{160} = \frac{1}{5}.$$

$$\boxed{\text{(a) } \frac{1}{5}}$$

Q4.25.1 NAT 2025 2M Ans: 3 ● ○ ○

For an irrotational flow,

$$\nabla \times \vec{V} = 0.$$

Given velocity field,

$$\vec{V} = 3z\,\hat{i} + 0\,\hat{j} + Cx\,\hat{k}.$$

Using curl formula,

$$\nabla \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ 3z & 0 & Cx \end{vmatrix}.$$

Only the $\hat{j}$ component is non-zero:

$$\nabla \times \vec{V} = \left(\frac{\partial(3z)}{\partial z} - \frac{\partial(Cx)}{\partial x}\right)\hat{j}.$$

$$\nabla \times \vec{V} = (3 - C)\,\hat{j}.$$

For irrotational flow,

$$3 - C = 0$$

$$C = 3.$$

$$\boxed{3}$$

Q4.25.2 NAT 2025 2M Ans: 1 ● ○ ○

Given

$$f(x,y) = x^2 + xy^2.$$

The directional derivative in the direction of unit vector $\hat{u}$ is

$$D_{\hat{u}} f = \nabla f \cdot \hat{u}.$$

First, find the gradient:

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right).$$

$$\frac{\partial f}{\partial x} = 2x + y^2, \qquad \frac{\partial f}{\partial y} = 2xy.$$

At the point $(1,0)$,

$$\nabla f(1,0) = (2,0).$$

The given direction vector is

$$\frac{1}{2}(\hat{\imath} + \sqrt{3}\hat{\jmath}) = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right),$$

which is already a unit vector.
Hence,

$$D_{\hat{u}} f = (2,0) \cdot \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right).$$

$$D_{\hat{u}} f = 1.$$

$$\boxed{1.0}$$

Q4.25.3 MCQ 2025 1M Ans: D ● ○ ○

For any continuously differentiable vector field $\vec{F}$, the vector calculus identity is

$$\nabla \cdot (\nabla \times \vec{F}) = 0.$$

Hence, the divergence of the curl of a vector field is always zero.

$$\boxed{\text{(d) Zero}}$$

Q4.24.1 MCQ 2024 1M Ans: B ● ● ○

Method 1

The given surface integral is

$$\iint_S z\,dx\,dy,$$

where S is the external surface of the sphere

$$x^2 + y^2 + z^2 = R^2.$$

Using the property of surface integrals,

$$\iint_S z\,dx\,dy = \iint_S z\,n_z\,dS,$$

where n_z is the z-component of the outward unit normal.
For the sphere,

$$n_z = \frac{z}{R}.$$

Hence,

$$\iint_S z\,dx\,dy = \frac{1}{R}\iint_S z^2\,dS.$$

By symmetry over the sphere,

$$\iint_S x^2\,dS = \iint_S y^2\,dS = \iint_S z^2\,dS.$$

Also,

$$x^2 + y^2 + z^2 = R^2.$$

Therefore,

$$3\iint_S z^2\,dS = \iint_S R^2\,dS = R^2(4\pi R^2).$$

$$\iint_S z^2\,dS = \frac{4\pi R^4}{3}.$$

Thus,

$$\iint_S z\,dx\,dy = \frac{1}{R}\cdot\frac{4\pi R^4}{3} = \frac{4\pi R^3}{3}.$$

Method 2

Let

$$\vec{F} = 0\,\hat{\imath} + 0\,\hat{\jmath} + z\,\hat{k}.$$

Comparing with the standard surface integral form,

$$\iint_S \vec{F}\cdot\hat{n}\,dS = \iint_S z\,dx\,dy.$$

Using the Divergence Theorem,

$$\iint_S \vec{F}\cdot\hat{n}\,dS = \iiint_V (\nabla\cdot\vec{F})\,dV.$$

Now,

$$\nabla\cdot\vec{F} = \frac{\partial}{\partial x}(0) + \frac{\partial}{\partial y}(0) + \frac{\partial}{\partial z}(z) = 1.$$

Therefore,

$$\iint_S z\,dx\,dy = \iiint_V 1\,dV = \iiint_V dV.$$

This equals the volume of the sphere of radius R:

$$\iiint_V dV = \frac{4}{3}\pi R^3.$$

$$\boxed{\text{(b) } \frac{4\pi R^3}{3}}$$

Q4.23.1 NAT 2023 1M Ans: 0 ● ○ ○

Given vector field

$$\vec{B}(x, y, z) = x\hat{i} + y\hat{j} - 2z\hat{k}.$$

We need to evaluate

$$\int_S \vec{B} \cdot \hat{n}\, dS,$$

where S is the curved surface of the cone.
Use the divergence theorem on the closed cone (curved surface + base).

$$\nabla \cdot \vec{B} = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial(-2z)}{\partial z} = 1 + 1 - 2 = 0.$$

Hence, total flux through the closed surface is

$$\iint_{\text{closed}} \vec{B} \cdot \hat{n}\, dS = 0.$$

Therefore,

$$\iint_S \vec{B} \cdot \hat{n}\, dS = -\iint_{\text{base}} \vec{B} \cdot \hat{n}\, dS.$$

For the base, outward normal is $-\hat{k}$ and $z = 0$. Hence,

$$\vec{B} = x\hat{i} + y\hat{j}$$

Thus,

$$\vec{B} \cdot (-\hat{k})dxdy = 0.$$

So flux through the base is zero, giving

$$\iint_S \vec{B} \cdot \hat{n}\, dS = 0.$$

$$\boxed{0}$$

Q4.22.1 MCQ 2022 1M Ans: A ● ○ ○

Given vector field

$$\vec{F} = 3x\hat{i} + 5y\hat{j} + 6z\hat{k}.$$

We need to evaluate the surface integral over the closed surface of the cube:

$$\iint_A \vec{F} \cdot d\vec{A}.$$

Using the divergence theorem,

$$\iint_A \vec{F} \cdot d\vec{A} = \iiint_V (\nabla \cdot \vec{F})\, dV.$$

Compute divergence:

$$\nabla \cdot \vec{F} = \frac{\partial(3x)}{\partial x} + \frac{\partial(5y)}{\partial y} + \frac{\partial(6z)}{\partial z}.$$

$$\nabla \cdot \vec{F} = 3 + 5 + 6 = 14.$$

The cube has unit edge length, so its volume is

$$V = 1.$$

Hence,

$$\iint_A \vec{F} \cdot d\vec{A} = 14 \times 1 = 14.$$

$$\boxed{\text{(a) } 14}$$

Q4.22.2 NAT 2022 2M Ans: 8.33 ● ● ○

Given vectors

$$\vec{a} = 5\hat{i} + 7\hat{j} + 2\hat{k}, \qquad \vec{b} = 3\hat{i} - \hat{j} + 6\hat{k}.$$

The component of $\vec{a}$ orthogonal to $\vec{b}$ is

$$\vec{a}_\perp = \vec{a} - \text{proj}_{\vec{b}}\vec{a}.$$

Its magnitude is

$$|\vec{a}_\perp| = \sqrt{|\vec{a}|^2 - \left(\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}\right)^2}.$$

Compute dot product:

$$\vec{a}\cdot\vec{b} = 5(3) + 7(-1) + 2(6) = 15 - 7 + 12 = 20.$$

Now,

$$|\vec{a}|^2 = 5^2 + 7^2 + 2^2 = 25 + 49 + 4 = 78,$$

and

$$|\vec{b}|^2 = 3^2 + (-1)^2 + 6^2 = 9 + 1 + 36 = 46.$$

Therefore,

$$|\vec{a}_\perp| = \sqrt{78 - \frac{20^2}{46}}.$$

$$|\vec{a}_\perp| = \sqrt{78 - \frac{400}{46}} = \sqrt{69.3043} \approx 8.325.$$

$$\boxed{8.33}$$

Q4.22.3 MCQ 2022 1M Ans: A ● ○ ○

Given

$$\phi = \frac{1}{2}(x^2 + y^2 + z^2).$$

First, compute the gradient:

$$\nabla\phi = \left(\frac{\partial\phi}{\partial x}, \frac{\partial\phi}{\partial y}, \frac{\partial\phi}{\partial z}\right) = x\hat{i} + y\hat{j} + z\hat{k}.$$

On the unit sphere,

$$x^2 + y^2 + z^2 = 1.$$

The outward unit normal vector is

$$\hat{n} = x\hat{i} + y\hat{j} + z\hat{k}.$$

Hence,

$$\hat{n}\cdot\nabla\phi = x^2 + y^2 + z^2 = 1.$$

Therefore,

$$\iint_S \hat{n}\cdot\nabla\phi\, dS = \iint_S 1\, dS.$$

This equals the surface area of the unit sphere:

$$\iint_S dS = 4\pi.$$

$$\boxed{\text{(a) } 4\pi}$$

Q4.20.1 MCQ 2020 2M Ans: A ● ○ ○

Given

$$f(x, y, z) = xyz.$$

The directional derivative in the direction of vector

$$\vec{v} = \hat{i} - 2\hat{j} + 2\hat{k}$$

is given by

$$D_{\hat{u}} f = \nabla f \cdot \hat{u},$$

where $\hat{u}$ is the unit vector along $\vec{v}$.
First, compute the gradient:

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right) = (yz, xz, xy).$$

At the point $(-1, 1, 3)$,

$$\nabla f = (-1 \times 3,\ -1 \times 3,\ -1 \times 1) = (3, -3, -1).$$

Magnitude of $\vec{v}$:

$$|\vec{v}| = \sqrt{1^2 + (-2)^2 + 2^2} = 3.$$

Hence, the unit vector is

$$\hat{u} = \left(\frac{1}{3}, -\frac{2}{3}, \frac{2}{3}\right).$$

Now,

$$D_{\hat{u}} f = (3, -3, -1) \cdot \left(\frac{1}{3}, -\frac{2}{3}, \frac{2}{3}\right).$$

$$D_{\hat{u}} f = 1 + 2 - \frac{2}{3} = \frac{7}{3}.$$

$$\boxed{\text{(a) } \frac{7}{3}}$$

Q4.20.2 MCQ 2020 2M Ans: A ● ○ ○

Given vector field

$$\vec{F}(x, y, z) = \frac{x}{(x^2 + y^2 + z^2)^{3/2}}\hat{i} + \frac{y}{(x^2 + y^2 + z^2)^{3/2}}\hat{j} + \frac{z}{(x^2 + y^2 + z^2)^{3/2}}\hat{k}.$$

This can be written as

$$\vec{F} = \frac{\vec{r}}{r^3} = \frac{\hat{r}}{r^2}.$$

For a sphere of radius r, the outward unit normal is $\hat{r}$.
Hence,

$$\vec{F} \cdot d\vec{S} = \frac{\hat{r}}{r^2} \cdot (\hat{r}\, dS) = \frac{dS}{r^2}.$$

Flux through outer sphere $(r = 2)$:

$$\iint \vec{F} \cdot d\vec{S} = \frac{1}{2^2} \times 4\pi(2^2) = 4\pi.$$

For the inner sphere $(r = 1)$, outward normal of the shell is directed toward the center, hence

$$d\vec{S} = -\hat{r}\, dS.$$

Therefore,

$$\iint \vec{F} \cdot d\vec{S} = -4\pi.$$

Net flux through both surfaces:

$$4\pi - 4\pi = 0.$$

$$\boxed{\text{(a) } 0}$$

Key Points

- The field $\frac{\vec{r}}{r^3}$ is radial and inversely proportional to r^2.
- Flux through any closed spherical surface centered at origin is 4π.
- Inner surface normal for a shell points inward toward the center.

Mistakes to be avoided

- Do not use the same outward normal direction for both spherical surfaces.
- Remember that the shell has two boundary surfaces with opposite orientations.

Q4.20.3 NAT 2020 1M Ans: 6 ● ○ ○

Given vectors

$$\vec{A} = 2\hat{j} - 3\hat{k}, \qquad \vec{B} = -2\hat{i} + \hat{k}, \qquad \vec{C} = 3\hat{i} - \hat{j}.$$

We need to evaluate

$$\vec{A} \cdot (\vec{B} \times \vec{C}) + 6.$$

First, compute $\vec{B} \times \vec{C}$:

$$\vec{B} \times \vec{C} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 0 & 1 \\ 3 & -1 & 0 \end{vmatrix}.$$

$$\vec{B} \times \vec{C} = \hat{i}(0 - (-1)) - \hat{j}(0 - 3) + \hat{k}(2)$$

$$\vec{B} \times \vec{C} = \hat{i} + 3\hat{j} + 2\hat{k}.$$

Now,

$$\vec{A} \cdot (\vec{B} \times \vec{C}) = (0, 2, -3) \cdot (1, 3, 2).$$

$$= 0 + 6 - 6 = 0.$$

Hence,

$$\vec{A} \cdot (\vec{B} \times \vec{C}) + 6 = 0 + 6 = 6.$$

$$\boxed{6}$$

Key Points

- $\vec{A} \cdot (\vec{B} \times \vec{C})$ is the scalar triple product.
- Scalar triple product gives the signed volume of the parallelepiped formed by the vectors.
- Evaluate cross product before taking dot product.

Q4.19.1 MCQ 2019 2M Ans: B ● ○ ○

Given

$$\vec{U} = \frac{1}{3}\left(-y^3\hat{\imath} + x^3\hat{\jmath} + z^3\hat{k}\right).$$

We need to evaluate

$$\iint_S (\nabla \times \vec{U}) \cdot \hat{n}\, dS,$$

over the curved surface of the hemisphere

$$x^2 + y^2 + z^2 = 1, \qquad z \geq 0.$$

Using Stokes' theorem,

$$\iint_S (\nabla \times \vec{U}) \cdot \hat{n}\, dS = \oint_C \vec{U} \cdot d\vec{r},$$

where C is the boundary circle

$$x^2 + y^2 = 1, \qquad z = 0.$$

On C,

$$\vec{U} = \frac{1}{3}\left(-y^3\hat{\imath} + x^3\hat{\jmath}\right).$$

Parameterize the circle:

$$x = \cos\theta, \qquad y = \sin\theta, \qquad 0 \leq \theta \leq 2\pi.$$

Then,

$$d\vec{r} = (-\sin\theta\,\hat{\imath} + \cos\theta\,\hat{\jmath})\,d\theta.$$

Hence,

$$\vec{U} \cdot d\vec{r} = \frac{1}{3}\left(\sin^4\theta + \cos^4\theta\right) d\theta.$$

Therefore,

$$\oint_C \vec{U} \cdot d\vec{r} = \frac{1}{3}\int_0^{2\pi} (\sin^4\theta + \cos^4\theta)\, d\theta.$$

Using

$$\int_0^{2\pi} \sin^4\theta\, d\theta = \int_0^{2\pi} \cos^4\theta\, d\theta = \frac{3\pi}{4},$$

we get

$$\oint_C \vec{U} \cdot d\vec{r} = \frac{1}{3}\left(\frac{3\pi}{4} + \frac{3\pi}{4}\right) = \frac{\pi}{2}.$$

$$\boxed{\text{(b) } \frac{\pi}{2}}$$

Q4.19.2 MCQ 2019 1M Ans: C ● ○ ○

Given

$$f(x, y) = x^2 + y^2.$$

The direction is from $(0, 0)$ to $(1, 1)$, so the direction vector is

$$\vec{v} = \hat{\imath} + \hat{\jmath}.$$

Its magnitude is

$$|\vec{v}| = \sqrt{1^2 + 1^2} = \sqrt{2}.$$

Hence, the unit vector is

$$\hat{u} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right).$$

The directional derivative is

$$D_{\hat{u}} f = \nabla f \cdot \hat{u}.$$

First, compute the gradient:

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right) = (2x, 2y).$$

At the point $(1, 1)$,

$$\nabla f(1, 1) = (2, 2).$$

Therefore,

$$D_{\hat{u}} f = (2, 2) \cdot \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right).$$

$$D_{\hat{u}} f = \frac{2}{\sqrt{2}} + \frac{2}{\sqrt{2}} = 2\sqrt{2}.$$

$$\boxed{\text{(c) } 2\sqrt{2}}$$

Q4.19.3 MCQ 2019 1M Ans: C ● ○ ○

The position vector of point $P(20, 10)$ is rotated anticlockwise by $30°$ about the origin. Using the rotation transformation,

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos 30° & -\sin 30° \\ \sin 30° & \cos 30° \end{bmatrix} \begin{bmatrix} 20 \\ 10 \end{bmatrix}$$

Applying the rotation matrix,

$$x = 20\cos 30° - 10\sin 30°$$

$$= 20\left(\frac{\sqrt{3}}{2}\right) - 10\left(\frac{1}{2}\right) = 10\sqrt{3} - 5 \approx 12.32$$

$$y = 20\sin 30° + 10\cos 30°$$

$$= 20\left(\frac{1}{2}\right) + 10\left(\frac{\sqrt{3}}{2}\right) = 10 + 5\sqrt{3} \approx 18.66$$

Hence,

$$Q = (12.32,\ 18.66)$$

$$\boxed{\text{(c) } (12.32,\ 18.66)}$$

Key Points

- Anticlockwise rotation by angle θ uses
$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$
- Rotation about the origin preserves the magnitude of the position vector.

Q4.18.1 MCQ 2018 2M Ans: C ● ○ ○

Given

$$\phi = \ln |\vec{r}|,$$

where

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}, \qquad |\vec{r}| = \sqrt{x^2 + y^2 + z^2}.$$

Using logarithm property,

$$\phi = \ln\left(\sqrt{x^2 + y^2 + z^2}\right) = \frac{1}{2}\ln(x^2 + y^2 + z^2).$$

Now compute the gradient:

$$\nabla\phi = \left(\frac{\partial\phi}{\partial x}, \frac{\partial\phi}{\partial y}, \frac{\partial\phi}{\partial z}\right).$$

$$\frac{\partial\phi}{\partial x} = \frac{x}{x^2 + y^2 + z^2}, \quad \frac{\partial\phi}{\partial y} = \frac{y}{x^2 + y^2 + z^2}, \quad \frac{\partial\phi}{\partial z} = \frac{z}{x^2 + y^2 + z^2}.$$

Hence,

$$\nabla\phi = \frac{x\hat{i} + y\hat{j} + z\hat{k}}{x^2 + y^2 + z^2}.$$

Since

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}, \qquad \vec{r}\cdot\vec{r} = x^2 + y^2 + z^2,$$

we get

$$\nabla\phi = \frac{\vec{r}}{\vec{r}\cdot\vec{r}}.$$

$$\boxed{(c)\ \frac{\vec{r}}{\vec{r}\cdot\vec{r}}}$$

Q4.18.2 MCQ 2018 1M Ans: C ● ○ ○

Given vector field,

$$\vec{u} = e^x(\cos y\,\hat{i} + \sin y\,\hat{j})$$

For a vector field,

$$\vec{F} = F_x\hat{i} + F_y\hat{j}$$

the divergence is

$$\nabla\cdot\vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y}$$

Here,

$$F_x = e^x\cos y, \qquad F_y = e^x\sin y$$

Now,

$$\frac{\partial}{\partial x}(e^x\cos y) = e^x\cos y$$

and

$$\frac{\partial}{\partial y}(e^x\sin y) = e^x\cos y$$

Therefore,

$$\nabla\cdot\vec{u} = e^x\cos y + e^x\cos y$$

$$\nabla\cdot\vec{u} = 2e^x\cos y$$

$$\boxed{(c)\ 2e^x\cos y}$$

Q4.18.3 MCQ 2018 2M Ans: C ● ○ ○

Using Divergence Theorem,

$$\oiint_S \vec{r}\cdot\vec{n}\,ds = \iiint_V (\nabla\cdot\vec{r})\,dV$$

Given,

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

Hence,

$$\nabla\cdot\vec{r} = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z}$$

$$\nabla\cdot\vec{r} = 1 + 1 + 1 = 3$$

Therefore,

$$\oiint_S \vec{r}\cdot\vec{n}\,ds = \iiint_V 3\,dV$$

$$= 3V$$

$$\boxed{\text{(c) } 3V}$$

Q4.17.1 NAT 2017 2M Ans: 226 ● ○ ○

Using Divergence Theorem,

$$\iint_S \vec{F}\cdot\hat{n}\,dS = \iiint_V (\nabla\cdot\vec{F})\,dV$$

Given,

$$\vec{F} = (x+y)\hat{i} + (x+z)\hat{j} + (y+z)\hat{k}$$

Hence,

$$\nabla\cdot\vec{F} = \frac{\partial(x+y)}{\partial x} + \frac{\partial(x+z)}{\partial y} + \frac{\partial(y+z)}{\partial z}$$

$$\nabla\cdot\vec{F} = 1 + 0 + 1 = 2$$

The surface is the sphere $x^2 + y^2 + z^2 = 9$ with radius $r = 3$
Volume of sphere:

$$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(3)^3 = 36\pi$$

Therefore,

$$\iint_S \vec{F}\cdot\hat{n}\,dS = \iiint_V 2\,dV = 2(36\pi) = 72\pi = 226.19$$

$$\boxed{226}$$

Q4.17.2 NAT 2017 2M Ans: 0 ● ○ ○

Given,

$$\vec{V} = 2yz\,\hat{i} + 3xz\,\hat{j} + 4xy\,\hat{k}$$

First calculate curl of $\vec{V}$:

$$\nabla\times\vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ 2yz & 3xz & 4xy \end{vmatrix}$$

$$\nabla \times \vec{V} = (4x - 3x)\hat{i} - (4y - 2y)\hat{j} + (3z - 2z)\hat{k}$$

$$\nabla \times \vec{V} = x\hat{i} - 2y\hat{j} + z\hat{k}$$

Now,

$$\nabla \cdot (\nabla \times \vec{V}) = \frac{\partial x}{\partial x} + \frac{\partial(-2y)}{\partial y} + \frac{\partial z}{\partial z}$$

$$= 1 - 2 + 1$$

$$\boxed{0}$$

Key Points

- Divergence of curl of any vector field is always zero.
- Identity:

$$\nabla \cdot (\nabla \times \vec{A}) = 0$$

Q4.17.3 MCQ 2017 2M Ans: C ● ○ ○

The given parametric curve is

$$x = \cos\left(\frac{\pi u}{2}\right), \qquad y = \sin\left(\frac{\pi u}{2}\right), \qquad 0 \le u \le 1$$

which represents the first-quadrant arc of the unit circle:

$$x^2 + y^2 = 1.$$

For rotation about the x-axis, surface area is

$$S = 2\pi \int y\, ds.$$

Now,

$$\frac{dx}{du} = -\frac{\pi}{2}\sin\left(\frac{\pi u}{2}\right), \qquad \frac{dy}{du} = \frac{\pi}{2}\cos\left(\frac{\pi u}{2}\right)$$

Hence,

$$ds = \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, du = \frac{\pi}{2}\, du.$$

Therefore,

$$S = 2\pi \int_0^1 \sin\left(\frac{\pi u}{2}\right) \frac{\pi}{2}\, du = \pi^2 \int_0^1 \sin\left(\frac{\pi u}{2}\right) du.$$

Let

$$t = \frac{\pi u}{2}, \qquad du = \frac{2}{\pi} dt.$$

Then

$$S = \pi^2 \cdot \frac{2}{\pi} \int_0^{\pi/2} \sin t\, dt = 2\pi \Big[-\cos t \Big]_0^{\pi/2} = 2\pi.$$

$$\boxed{\text{(c) } 2\pi}$$

Key Points

- Surface area generated by revolving a curve about the x-axis is

$$S = 2\pi \int y\, ds.$$

Q4.17.4 MCQ 2017 1M Ans: B ● ○ ○

For a two-dimensional velocity field

$$\vec{V} = u\,\hat{i} + v\,\hat{j}$$

where

$$u = 5 + a_1 x + b_1 y, \qquad v = 4 + a_2 x + b_2 y,$$

the condition for incompressible flow is

$$\nabla \cdot \vec{V} = 0.$$

Hence,

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0.$$

Calculating the derivatives,

$$\frac{\partial u}{\partial x} = a_1, \qquad \frac{\partial v}{\partial y} = b_2.$$

Therefore,

$$a_1 + b_2 = 0.$$

$$\boxed{\text{(b) } a_1 + b_2 = 0}$$

Key Points

- Incompressible flow satisfies the continuity equation

$$\nabla \cdot \vec{V} = 0.$$

Q4.16.1 NAT 2016 2M Ans: 16 ● ○ ○

Using Green's theorem,

$$\oint_C \vec{F} \cdot \vec{r}\, ds = \oint_C (M\, dx + N\, dy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA$$

Given,

$$\vec{F}(x, y) = y\hat{i} + 2x\hat{j}$$

Hence,

$$M = y, \qquad N = 2x$$

Therefore,

$$\frac{\partial N}{\partial x} = 2, \qquad \frac{\partial M}{\partial y} = 1$$

$$\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 1$$

So,

$$\oint_C \vec{F} \cdot \vec{r}\, ds = \iint_R 1\, dA$$

Radius of circle:

$$r = \frac{4}{\sqrt{\pi}}$$

Area of circle:

$$A = \pi r^2 = \pi \left(\frac{4}{\sqrt{\pi}}\right)^2 = 16$$

Hence,

$$\oint_C \vec{F} \cdot \vec{r}\, ds = 16$$

$$\boxed{16}$$

Q4.16.2 NAT | 2016 | 2M | Ans: 726 | ● ○ ○

Given,

$$\nabla\phi = yz\,\hat{i} + xz\,\hat{j} + xy\,\hat{k}$$

Since the vector field is a gradient field,

$$\int_C \nabla\phi \cdot d\vec{r} = \phi(B) - \phi(A)$$

First determine the scalar potential ϕ.

$$\frac{\partial\phi}{\partial x} = yz$$

Integrating with respect to x,

$$\phi = xyz + f(y, z)$$

Now,

$$\frac{\partial\phi}{\partial y} = xz$$

which matches the given field, hence

$$f(y, z) = g(z)$$

Also,

$$\frac{\partial\phi}{\partial z} = xy$$

therefore,

$$g'(z) = 0$$

Hence,

$$\phi = xyz$$

The curve is parameterized by

$$x = t, \qquad y = t^2, \qquad z = 3t^2$$

For $t = 1$:

$$A = (1, 1, 3)$$

For $t = 3$:

$$B = (3, 9, 27)$$

Now,

$$\phi(B) = 3 \cdot 9 \cdot 27 = 729$$

$$\phi(A) = 1 \cdot 1 \cdot 3 = 3$$

Therefore,

$$\int_C \nabla\phi \cdot d\vec{r} = 729 - 3 = 726$$

$$\boxed{726}$$

Q4.15.1 NAT 2015 2M Ans: 216 ● ○ ○

Using Divergence Theorem,

$$\iint_S \frac{1}{\pi}(9x\hat{i} - 3y\hat{j}) \cdot \hat{n}\, dS = \iiint_V \nabla \cdot \left(\frac{1}{\pi}(9x\hat{i} - 3y\hat{j})\right) dV$$

Given vector field,

$$\vec{F} = \frac{1}{\pi}(9x\hat{i} - 3y\hat{j})$$

Hence,

$$\nabla \cdot \vec{F} = \frac{1}{\pi}\left(\frac{\partial(9x)}{\partial x} + \frac{\partial(-3y)}{\partial y}\right)$$

$$\nabla \cdot \vec{F} = \frac{1}{\pi}(9 - 3) = \frac{6}{\pi}$$

The sphere is

$$x^2 + y^2 + z^2 = 9$$

with radius

$$r = 3$$

Volume of sphere:

$$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(27) = 36\pi$$

Therefore,

$$\iint_S \vec{F} \cdot \hat{n}\, dS = \frac{6}{\pi} \times 36\pi$$

$$= 216$$

$$\boxed{216}$$

Q4.15.2 NAT 2015 2M Ans: 1.67 ● ○ ○

Using Green's theorem,

$$\oint_C (M\,dx + N\,dy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dA$$

Given,

$$M = 3x - 8y^2, \qquad N = 4y - 6xy$$

Now,

$$\frac{\partial N}{\partial x} = -6y$$

and

$$\frac{\partial M}{\partial y} = -16y$$

Therefore,

$$\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = -6y + 16y = 10y$$

The region is bounded by

$$x = 0, \qquad y = 0, \qquad x + y = 1$$

Hence,

$$\iint_R 10y\, dA = \int_0^1 \int_0^{1-y} 10y\, dx\, dy$$

$$= \int_0^1 10y(1-y)\, dy$$

$$= 10 \int_0^1 (y - y^2)\, dy$$

$$= 10 \left[\frac{y^2}{2} - \frac{y^3}{3} \right]_0^1$$

$$= 10 \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{5}{3} = 1.67$$

$$\boxed{1.67}$$

Q4.15.3 MCQ 2015 1M Ans: A ● ○ ○

Given,

$$\vec{V}(x, y, z) = 2x^2\hat{i} + 3z^2\hat{j} + y^3\hat{k}$$

Curl of a vector field is

$$\nabla \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x^2 & 3z^2 & y^3 \end{vmatrix}$$

Expanding,

$$\nabla \times \vec{V} = \left(\frac{\partial y^3}{\partial y} - \frac{\partial 3z^2}{\partial z} \right)\hat{i} - \left(\frac{\partial y^3}{\partial x} - \frac{\partial 2x^2}{\partial z} \right)\hat{j} + \left(\frac{\partial 3z^2}{\partial x} - \frac{\partial 2x^2}{\partial y} \right)\hat{k}$$

$$= (3y^2 - 6z)\hat{i} - 0\hat{j} + 0\hat{k}$$

At

$$x = y = z = 1$$

$$\nabla \times \vec{V} = (3 - 6)\hat{i} = -3\hat{i}$$

$$\boxed{\text{(a) } -3\hat{i}}$$

Q4.15.4 NAT 2015 2M Ans: 1.86 ● ○ ○

$$L = \int_0^{\pi/2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt$$

Given,

$$x(t) = \cos t, \qquad y(t) = \sin t, \qquad z(t) = \frac{2}{\pi}t$$

Differentiating,

$$\frac{dx}{dt} = -\sin t, \qquad \frac{dy}{dt} = \cos t, \qquad \frac{dz}{dt} = \frac{2}{\pi}$$

Hence,

$$L = \int_0^{\pi/2} \sqrt{\sin^2 t + \cos^2 t + \left(\frac{2}{\pi}\right)^2}\, dt$$

$$= \int_0^{\pi/2} \sqrt{1 + \frac{4}{\pi^2}}\, dt = \frac{\pi}{2}\sqrt{1 + \frac{4}{\pi^2}}$$

$$= \frac{\pi}{2} \cdot \frac{\sqrt{\pi^2 + 4}}{\pi} = \frac{\sqrt{\pi^2 + 4}}{2} = 1.86$$

$$\boxed{1.86}$$

Key Points

- Arc length of a space curve is

$$L = \int \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt.$$

Mistakes to be avoided

- Do not use the 2D arc length formula; this is a space curve.

Q4.14.1 MCQ 2014 1M Ans: C ● ○ ○

Given vector field,

$$\vec{F} = x^2 z\,\hat{\imath} + xy\,\hat{\jmath} - yz^2\,\hat{k}$$

Divergence of a vector field is

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(x^2 z) + \frac{\partial}{\partial y}(xy) + \frac{\partial}{\partial z}(-yz^2)$$

Calculating each term,

$$\frac{\partial}{\partial x}(x^2 z) = 2xz$$

$$\frac{\partial}{\partial y}(xy) = x$$

$$\frac{\partial}{\partial z}(-yz^2) = -2yz$$

Therefore,

$$\nabla \cdot \vec{F} = 2xz + x - 2yz$$

At point

$$(1, -1, 1)$$

$$\nabla \cdot \vec{F} = 2(1)(1) + 1 - 2(-1)(1)$$

$$= 2 + 1 + 2 = 5$$

$$\boxed{(c)\ 5}$$

Q4.14.2 MCQ 2014 1M Ans: A ● ○ ○

For a vector field

$$\vec{F} = x^2z^2\hat{i} - 2xy^2z\hat{j} + 2y^2z^3\hat{k}$$

the curl is given by

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2z^2 & -2xy^2z & 2y^2z^3 \end{vmatrix}$$

Expanding the determinant,

$$\nabla \times \vec{F} = \hat{i}\left(\frac{\partial}{\partial y}(2y^2z^3) - \frac{\partial}{\partial z}(-2xy^2z)\right)$$

$$-\hat{j}\left(\frac{\partial}{\partial x}(2y^2z^3) - \frac{\partial}{\partial z}(x^2z^2)\right) + \hat{k}\left(\frac{\partial}{\partial x}(-2xy^2z) - \frac{\partial}{\partial y}(x^2z^2)\right)$$

Now,

$$\frac{\partial}{\partial y}(2y^2z^3) = 4yz^3, \qquad \frac{\partial}{\partial z}(-2xy^2z) = -2xy^2$$

$$\frac{\partial}{\partial x}(2y^2z^3) = 0, \qquad \frac{\partial}{\partial z}(x^2z^2) = 2x^2z$$

$$\frac{\partial}{\partial x}(-2xy^2z) = -2y^2z, \qquad \frac{\partial}{\partial y}(x^2z^2) = 0$$

Therefore,

$$\nabla \times \vec{F} = (4yz^3 + 2xy^2)\hat{i} - (-2x^2z)\hat{j} - 2y^2z\hat{k}$$

$$\nabla \times \vec{F} = (4yz^3 + 2xy^2)\hat{i} + 2x^2z\hat{j} - 2y^2z\hat{k}$$

$$\boxed{\text{(a) } (4yz^3 + 2xy^2)\hat{i} + 2x^2z\hat{j} - 2y^2z\hat{k}}$$

Q4.14.3 MCQ 2014 1M Ans: B ● ○ ○

Let

$$\vec{A} = \hat{i} + \hat{j} + \hat{k}, \qquad \vec{B} = 2\hat{i} + 3\hat{j} + \hat{k}, \qquad \vec{C} = 5\hat{i} + 6\hat{j} + 4\hat{k}$$

To check whether the vectors are linearly dependent or independent, form the determinant:

$$\begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 5 & 6 & 4 \end{vmatrix}$$

Expanding,

$$= 1(12 - 6) - 1(8 - 5) + 1(12 - 15)$$

$$= 6 - 3 - 3$$

$$= 0$$

Since the determinant is zero, the vectors are linearly dependent.

$$\boxed{\text{(b) The vectors are linearly dependent}}$$

Key Points

- Three vectors are linearly dependent if the determinant formed by their components is zero.
- Non-zero determinant implies linear independence.
- Mutually perpendicular vectors must have zero dot product pairwise.

Q4.14.4 MCQ 2014 2M Ans: C ● ● ○

Method 1

Using Green's theorem,

$$\oint_C (y\,dx - x\,dy) = \iint_A \left(\frac{\partial(-x)}{\partial x} - \frac{\partial y}{\partial y} \right) dA$$
$$= \iint_A (-1-1)\,dA = -2 \iint_A dA.$$

The curve is the circle

$$x^2 + y^2 = \frac{1}{4},$$

whose radius is

$$r = \frac{1}{2}.$$

Hence,

$$\text{Area}(A) = \pi r^2 = \pi \left(\frac{1}{2}\right)^2 = \frac{\pi}{4}.$$

Therefore,

$$\oint_C (y\,dx - x\,dy) = -2\left(\frac{\pi}{4}\right) = -\frac{\pi}{2}.$$

Method 2

Parameterize the circle as

$$x = \frac{1}{2}\cos\theta, \qquad y = \frac{1}{2}\sin\theta, \qquad 0 \le \theta \le 2\pi.$$

Then

$$dx = -\frac{1}{2}\sin\theta\,d\theta, \qquad dy = \frac{1}{2}\cos\theta\,d\theta.$$

Thus,

$$y\,dx - x\,dy = -\frac{1}{4}(\sin^2\theta + \cos^2\theta)\,d\theta = -\frac{1}{4}\,d\theta.$$

Hence,

$$\oint_C (y\,dx - x\,dy) = -\frac{1}{4}\int_0^{2\pi} d\theta = -\frac{\pi}{2}.$$

$$\boxed{\text{(c) } -\frac{\pi}{2}}$$

Q4.13.1 MCQ 2013 2M Ans: A ● ○ ○

Given,

$$\vec{F} = x\hat{i} + y\hat{j} + z\hat{k}$$

and the surface integral

$$\iint_S \frac{1}{4}(\vec{F}\cdot\vec{n})\, dA$$

where S is the sphere

$$x^2 + y^2 + z^2 = 1$$

Using Divergence Theorem,

$$\iint_S (\vec{F}\cdot\vec{n})\, dA = \iiint_V (\nabla\cdot\vec{F})\, dV$$

Now,

$$\nabla\cdot\vec{F} = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 1 + 1 + 1 = 3$$

Hence,

$$\iint_S \frac{1}{4}(\vec{F}\cdot\vec{n})\, dA = \frac{1}{4}\iiint_V 3\, dV$$

Volume of unit sphere:

$$V = \frac{4}{3}\pi$$

Therefore,

$$\frac{1}{4}\times 3\times\frac{4}{3}\pi = \pi$$

$$\boxed{\text{(a) } \pi}$$

Q4.12.1 MCQ 2012 1M Ans: A ● ○ ○

For the sphere

$$x^2 + y^2 + z^2 = 1$$

the outward unit normal vector at any point is directed along the radius vector.
At the point

$$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)$$

the position vector is

$$\vec{r} = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$$

Its magnitude is

$$|\vec{r}| = \sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2} = \sqrt{\frac{1}{2} + \frac{1}{2}} = 1$$

Hence, the vector is already a unit vector. Therefore, the outward unit normal vector is

$$\hat{n} = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$$

$$\boxed{\text{(a) } \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}}$$

Q4.10.1 MCQ 2010 2M Ans: D ● ○ ○

Given velocity field,

$$\vec{V} = 2xy\,\hat{i} - x^2z\,\hat{j}$$

The vorticity vector is

$$\vec{\omega} = \nabla \times \vec{V}$$

Here,

$$\vec{V} = (2xy)\hat{i} + (-x^2z)\hat{j} + 0\hat{k}$$

Using curl formula,

$$\vec{\omega} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy & -x^2z & 0 \end{vmatrix}$$

$$\vec{\omega} = \left(0 - \frac{\partial(-x^2z)}{\partial z}\right)\hat{i} + \left(\frac{\partial(2xy)}{\partial z} - 0\right)\hat{j} + \left(\frac{\partial(-x^2z)}{\partial x} - \frac{\partial(2xy)}{\partial y}\right)\hat{k}$$

$$\vec{\omega} = x^2\hat{i} + 0\hat{j} + (-2xz - 2x)\hat{k}$$

At point $(1, 1, 1)$,

$$\vec{\omega} = \hat{i} - 4\hat{k}$$

$$\boxed{\text{(d) } \hat{i} - 4\hat{k}}$$

Key Points

- Vorticity is related to the curl of the velocity field.

Q4.09.1 MCQ 2009 1M Ans: C ● ○ ○

Given vector field,

$$\vec{F} = 3xz\hat{i} + 2xy\hat{j} - yz^2\hat{k}$$

The divergence of a vector field is

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(3xz) + \frac{\partial}{\partial y}(2xy) + \frac{\partial}{\partial z}(-yz^2)$$

Calculating each term,

$$\frac{\partial}{\partial x}(3xz) = 3z$$
$$\frac{\partial}{\partial y}(2xy) = 2x$$
$$\frac{\partial}{\partial z}(-yz^2) = -2yz$$

Hence,

$$\nabla \cdot \vec{F} = 3z + 2x - 2yz$$

At the point $(1, 1, 1)$,

$$\nabla \cdot \vec{F} = 3(1) + 2(1) - 2(1)(1)$$

$$= 3 + 2 - 2 = 3$$

$$\boxed{\text{(c) } 3}$$

Q4.09.2 MCQ | 2009 | 2M | Ans: B | ● ● ○

The path AB is a quarter circle of unit radius centered at the origin.
Parameterize the path using

$$x = \cos\theta, \qquad y = \sin\theta$$

where the motion is counter-clockwise from

$$A(1,0) \to B(0,1)$$

Hence,

$$0 \le \theta \le \frac{\pi}{2}$$

The integral is

$$\int_{AB} (x+y)^2 \, ds$$

For unit circle,

$$ds = d\theta$$

Now,

$$(x+y)^2 = (\cos\theta + \sin\theta)^2$$

$$= \cos^2\theta + \sin^2\theta + 2\sin\theta\cos\theta$$

$$= 1 + \sin 2\theta$$

Therefore,

$$\int_0^{\pi/2} (1 + \sin 2\theta) \, d\theta$$

$$= \left[\theta - \frac{\cos 2\theta}{2}\right]_0^{\pi/2}$$

$$= \left(\frac{\pi}{2} + \frac{1}{2}\right) - \left(0 - \frac{1}{2}\right)$$

$$= \frac{\pi}{2} + 1$$

$$\boxed{\text{(b) } \frac{\pi}{2} + 1}$$

Key Points

- A quarter circle of unit radius is conveniently parameterized using trigonometric functions.
- For a unit circle, arc length element is $ds = d\theta$.
- Use trigonometric identity:

$$\sin^2\theta + \cos^2\theta = 1$$

Mistakes to be avoided

- Do not forget to replace ds correctly while parameterizing.
- Limits must follow the direction of traversal from A to B.

Q4.08.1 MCQ 2008 2M Ans: B ● ○ ○

Given scalar function

$$f(x, y, z) = x^2 + 2y^2 + z$$

The directional derivative in the direction of vector $\vec{a}$ is

$$\nabla f \cdot \hat{a}$$

First, calculate the gradient:

$$\nabla f = \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} + \frac{\partial f}{\partial z}\hat{k}$$

$$\nabla f = 2x\hat{i} + 4y\hat{j} + \hat{k}$$

At the point $P(1, 1, 2)$,

$$\nabla f = 2\hat{i} + 4\hat{j} + \hat{k}$$

Given direction vector,

$$\vec{a} = 3\hat{i} - 4\hat{j}$$

Magnitude of $\vec{a}$:

$$|\vec{a}| = \sqrt{3^2 + (-4)^2} = 5$$

Hence, unit vector along $\vec{a}$ is

$$\hat{a} = \frac{3}{5}\hat{i} - \frac{4}{5}\hat{j}$$

Therefore,

$$\nabla f \cdot \hat{a} = (2\hat{i} + 4\hat{j} + \hat{k}) \cdot \left(\frac{3}{5}\hat{i} - \frac{4}{5}\hat{j}\right)$$

$$= \frac{6}{5} - \frac{16}{5} = -\frac{10}{5} = -2$$

$$\boxed{\text{(b) -2}}$$

Q4.07.1 MCQ 2007 2M Ans: B ● ○ ○

Let the position vectors of the three vertices of the triangle be

$$\vec{a},\ \vec{b},\ \vec{c}$$

Two sides of the triangle can be written as

$$\vec{AB} = \vec{b} - \vec{a}$$

and

$$\vec{AC} = \vec{c} - \vec{a}$$

Area of a triangle formed by two vectors is

$$\text{Area} = \frac{1}{2}\left|\vec{AB} \times \vec{AC}\right|$$

Substituting,

$$\text{Area} = \frac{1}{2}\left|(\vec{b} - \vec{a}) \times (\vec{c} - \vec{a})\right|$$

Using the property

$$(\vec{b} - \vec{a}) \times (\vec{c} - \vec{a}) = (\vec{a} - \vec{b}) \times (\vec{a} - \vec{c})$$

Hence,

$$\text{Area} = \frac{1}{2}\left|(\vec{a} - \vec{b}) \times (\vec{a} - \vec{c})\right|$$

$$\boxed{\text{(b) } \frac{1}{2}\left|(\vec{a} - \vec{b}) \times (\vec{a} - \vec{c})\right|}$$

Key Points

- Magnitude of cross product gives the area of the parallelogram formed by two vectors.
- Area of triangle is half the area of the corresponding parallelogram.
- Dot product cannot represent area since it gives a scalar projection quantity.

Mistakes to be avoided

- Do not forget the modulus sign in the cross product expression.
- Triple scalar product gives volume, not area.

Q4.05.1 MCQ 2005 2M Ans: A ● ● ○

Given vector field,

$$\vec{V}(\vec{r}) = 2xyz\hat{i} + x^2z\hat{j} + x^2y\hat{k}$$

To check whether the line integral is path independent, verify if the field is conservative.
Let

$$\vec{V} = P\hat{i} + Q\hat{j} + R\hat{k}$$

where

$$P = 2xyz, \qquad Q = x^2z, \qquad R = x^2y$$

Now,

$$\frac{\partial P}{\partial y} = 2xz, \qquad \frac{\partial Q}{\partial x} = 2xz$$

$$\frac{\partial P}{\partial z} = 2xy, \qquad \frac{\partial R}{\partial x} = 2xy$$

$$\frac{\partial Q}{\partial z} = x^2, \qquad \frac{\partial R}{\partial y} = x^2$$

Hence,

$$\nabla \times \vec{V} = 0$$

Therefore, the field is conservative and the line integral is path independent.
Find the potential function ϕ such that

$$\nabla\phi = \vec{V}$$

Integrating,

$$\frac{\partial \phi}{\partial x} = 2xyz$$

$$\phi = x^2yz + C$$

Thus,

$$\int \vec{V} \cdot d\vec{r} = \phi(1,1,1) - \phi(0,0,0)$$

$$= (1^2 \cdot 1 \cdot 1) - 0 = 1$$

$$\boxed{\text{(a) } 1}$$

Q4.05.2 MCQ 2005 1M Ans: A ● ○ ○

For a change of variables in a double integral,

$$dx\,dy = \left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv.$$

Given,

$$x = uv, \qquad y = \frac{v}{u}.$$

The Jacobian is

$$\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} v & u \\ -\frac{v}{u^2} & \frac{1}{u} \end{vmatrix}.$$

$$J = v\left(\frac{1}{u}\right) - u\left(-\frac{v}{u^2}\right) = \frac{v}{u} + \frac{v}{u} = \frac{2v}{u}.$$

Hence,

$$f(x,y)\,dx\,dy = f\left(uv, \frac{v}{u}\right)\left|\frac{2v}{u}\right| du\,dv.$$

Therefore,

$$\phi(u,v) = \frac{2v}{u}.$$

$$\boxed{\text{(a) } \frac{2v}{u}}$$

Q4.04.1 MCQ 2004 1M Ans: C ● ○ ○

Given unit vectors

$$\vec{P} = (0.866,\ 0.500,\ 0)$$

and

$$\vec{Q} = (0.259,\ 0.966,\ 0)$$

The angle between two vectors is obtained using

$$\vec{P}\cdot\vec{Q} = |\vec{P}||\vec{Q}|\cos\theta$$

Since both are unit vectors,

$$|\vec{P}| = |\vec{Q}| = 1$$

Hence,

$$\cos\theta = (0.866)(0.259) + (0.500)(0.966)$$

$$= 0.224 + 0.483$$

$$\approx 0.707$$

Therefore,

$$\cos\theta = \frac{1}{\sqrt{2}}$$

$$\theta = 45^\circ$$

$$\boxed{\text{(c) } 45^\circ}$$

Q4.99.1 MCQ 1999 1M Ans: D ● ○ ○

For an ideal fluid, the velocity potential function ϕ must satisfy Laplace's equation:

$$\nabla^2\phi = 0$$

Given,

$$\phi = ax^2y - y^3$$

Applying Laplace operator,

$$\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} = 0$$

First, differentiate with respect to x:

$$\frac{\partial\phi}{\partial x} = 2axy$$

$$\frac{\partial^2\phi}{\partial x^2} = 2ay$$

Now, differentiate with respect to y:

$$\frac{\partial\phi}{\partial y} = ax^2 - 3y^2$$

$$\frac{\partial^2\phi}{\partial y^2} = -6y$$

Therefore,

$$2ay - 6y = 0$$

$$y(2a - 6) = 0$$

$$2a - 6 = 0$$

$$a = 3$$

$$\boxed{\text{(d) } 3}$$

Key Points

- Velocity potential for incompressible ideal fluid satisfies Laplace equation.
- Laplace equation is given by $\nabla^2\phi = 0$.
- Sum of second partial derivatives must be zero.

Q4.95.1 MCQ 1995 1M Ans: A ● ○ ○

Using the vector identity for curl of a scalar multiplied by a vector,

$$\nabla \times (f\vec{V}) = (\nabla f) \times \vec{V} + f(\nabla \times \vec{V})$$

Hence,

$$curl(f\vec{V}) = (gradf) \times \vec{V} + (fcurl\vec{V})$$

$$\boxed{\text{(a) } curl(f\vec{V}) = (gradf) \times \vec{V} + (fcurl\vec{V})}$$

Q4.26.1 MCQ 2026 2M Ans: A ● ○ ○

Given vector field,

$$\vec{V} = 3x^2yz\,\hat{\imath} - 5xy\,\hat{\jmath} + 6yz^2\,\hat{k}$$

Curl of a vector field is

$$\nabla \times \vec{V} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ 3x^2yz & -5xy & 6yz^2 \end{vmatrix}$$

$$\nabla \times \vec{V} = \left(\frac{\partial(6yz^2)}{\partial y} - \frac{\partial(-5xy)}{\partial z}\right)\hat{\imath} + \left(\frac{\partial(3x^2yz)}{\partial z} - \frac{\partial(6yz^2)}{\partial x}\right)\hat{\jmath} + \left(\frac{\partial(-5xy)}{\partial x} - \frac{\partial(3x^2yz)}{\partial y}\right)\hat{k}$$

$$\nabla \times \vec{V} = 6z^2\,\hat{\imath} + 3x^2y\,\hat{\jmath} + (-5y - 3x^2z)\,\hat{k}$$

At point $(2, -1, 1)$,

$$\nabla \times \vec{V} = 6(1)^2\,\hat{\imath} + 3(2)^2(-1)\,\hat{\jmath} + \left[-5(-1) - 3(2)^2(1)\right]\hat{k}$$

$$\nabla \times \vec{V} = 6\hat{\imath} - 12\hat{\jmath} - 7\hat{k}$$

$$\boxed{\text{(a) } 6\hat{\imath} - 12\hat{\jmath} - 7\hat{k}}$$

Q4.25.1 MSQ 2025 1M Ans: A,C ● ○ ○

Given velocity field,

$$\vec{V} = u\hat{x} + v\hat{y}$$

For a two-dimensional velocity field, curl is

$$\nabla \times \vec{V} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ u & v & 0 \end{vmatrix}$$

Hence, only the z-component exists:

$$\nabla \times \vec{V} = \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)\hat{z}$$

So, option A is correct.
Now, divergence of velocity is

$$\nabla \cdot \vec{V} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}$$

Thus, option C is correct.
Options B and D have incorrect expressions.

$$\boxed{\begin{array}{l} \text{(a)} \nabla \times \vec{V} = \left(\dfrac{\partial v}{\partial x} - \dfrac{\partial u}{\partial y}\right)\hat{z} \\ \text{(c) } \nabla \cdot \vec{V} = \dfrac{\partial u}{\partial x} + \dfrac{\partial v}{\partial y} \end{array}}$$

Q4.24.1 MSQ | 2024 | 2M | Ans: B,C,D | ● ● ○

Check each option separately.
Option A:

$$\vec{p} \times (\vec{q} \times \vec{r}) \stackrel{?}{=} (\vec{p} \times \vec{q}) \times \vec{r}$$

Cross product is *not associative*. Hence, in general,

$$\vec{p} \times (\vec{q} \times \vec{r}) \neq (\vec{p} \times \vec{q}) \times \vec{r}.$$

Therefore, option A is incorrect.
Option B:

$$\vec{r} \cdot (\vec{p} \times \vec{q}) \stackrel{?}{=} (\vec{q} \times \vec{p}) \cdot \vec{r}$$

Since

$$\vec{q} \times \vec{p} = -(\vec{p} \times \vec{q}),$$

the given equality is *not* a general vector identity. It would require

$$\vec{r} \cdot (\vec{p} \times \vec{q}) = 0.$$

For the given vectors,

$$\vec{p} = (1, 1, 1), \qquad \vec{q} = (1, 2, 3), \qquad \vec{r} = (2, 3, 4),$$

$$\vec{p} \times \vec{q} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & 2 & 3 \end{vmatrix} = \hat{i} - 2\hat{j} + \hat{k}.$$

Hence,

$$\vec{r} \cdot (\vec{p} \times \vec{q}) = (2, 3, 4) \cdot (1, -2, 1) = 2 - 6 + 4 = 0.$$

Therefore,

$$(\vec{q} \times \vec{p}) \cdot \vec{r} = -\vec{r} \cdot (\vec{p} \times \vec{q}) = 0.$$

Thus both sides are zero and option B is correct for the *given vectors*.
Option C:

$$\vec{p} \times (\vec{q} \times \vec{r}) + \vec{q} \times (\vec{r} \times \vec{p}) + \vec{r} \times (\vec{p} \times \vec{q}) = \vec{0}$$

This is the Jacobi identity of vector products.
Hence, option C is correct.
Option D: Using the vector triple product identity,

$$\vec{p} \times (\vec{q} \times \vec{r}) = (\vec{p} \cdot \vec{r})\vec{q} - (\vec{p} \cdot \vec{q})\vec{r}.$$

Hence, option D is correct.

$$\boxed{\begin{array}{l} \text{(b) } \vec{r} \cdot (\vec{p} \times \vec{q}) = (\vec{q} \times \vec{p}) \cdot \vec{r} \\ \text{(c) } \vec{p} \times (\vec{q} \times \vec{r}) + \vec{q} \times (\vec{r} \times \vec{p}) + \vec{r} \times (\vec{p} \times \vec{q}) = \vec{0} \\ \text{(d) } \vec{p} \times (\vec{q} \times \vec{r}) = (\vec{p} \cdot \vec{r})\vec{q} - (\vec{p} \cdot \vec{q})\vec{r} \end{array}}$$

Q4.24.2 MCQ 2024 2M Ans: A ● ○ ○

Given scalar field,

$$r = 6x^2 + 4y^2 - z^2 - 9xyz - 2xy + 3xz - yz$$

and vector field,

$$\vec{p} = (2x^2 - 3xy + z^2)\hat{\imath} + (2y^2 - 3yz + x^2)\hat{\jmath} + (2z^2 - 3zx + x^2)\hat{k}$$

For statement P:
A standard vector identity states that curl of gradient of any scalar field is always zero.

$$\nabla \times (\nabla r) = \vec{0}$$

Hence, statement P is TRUE.
For statement Q:
Another standard vector identity states that divergence of curl of any vector field is always zero.

$$\nabla \cdot (\nabla \times \vec{p}) = 0$$

Hence, statement Q is also TRUE.

$$\boxed{\text{(a) Both P and Q are TRUE}}$$

Mistakes to be avoided

- Do not attempt lengthy differentiation when standard identities apply.
- Curl and divergence operators are not interchangeable.

Q4.23.1 MCQ 2023 1M Ans: A ● ○ ○

Using the vector calculus identity for divergence of a scalar multiplied by a vector,

$$\nabla \cdot (\phi\vec{u}) = \phi(\nabla \cdot \vec{u}) + \vec{u} \cdot (\nabla\phi)$$

Hence,

$$div(\phi\vec{u}) = \phi\, div(\vec{u}) + \vec{u} \cdot grad(\phi)$$

$$\boxed{\text{(a) } div(\phi\vec{u}) = \phi\, div(\vec{u}) + \vec{u} \cdot grad(\phi)}$$

Q4.21.1 MCQ 2021 1M Ans: A ● ○ ○

Given surface,

$$x^2 + y^2 + z^2 - 48 = 0$$

Let,

$$F(x, y, z) = x^2 + y^2 + z^2 - 48$$

The normal vector to the surface is given by gradient of F:

$$\nabla F = \left(\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z}\right)$$

$$\nabla F = (2x, 2y, 2z)$$

At point $(4, 4, 4)$,

$$\nabla F = (8, 8, 8)$$

Magnitude of the vector is

$$|\nabla F| = \sqrt{8^2 + 8^2 + 8^2} = 8\sqrt{3}$$

Hence, the unit normal vector is

$$\left(\frac{8}{8\sqrt{3}}, \frac{8}{8\sqrt{3}}, \frac{8}{8\sqrt{3}}\right) = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$$

$$\boxed{\text{(a)} \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)}$$

Q4.21.2 NAT 2021 2M Ans: 1 ● ○ ○

Given function,

$$f(x, y) = xe^y$$

Directional derivative is given by

$$D_{\hat{u}} f = \nabla f \cdot \hat{u}$$

First, find gradient of f:

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right) = (e^y, xe^y)$$

At point $(2, 0)$,

$$\nabla f(2, 0) = (1, 2)$$

Direction vector from $(2, 0)$ to $\left(\frac{1}{2}, 2\right)$ is

$$\vec{d} = \left(\frac{1}{2} - 2, 2 - 0\right) = \left(-\frac{3}{2}, 2\right)$$

Magnitude of $\vec{d}$ is

$$|\vec{d}| = \sqrt{\left(-\frac{3}{2}\right)^2 + 2^2} = \frac{5}{2}$$

Unit vector along the direction is

$$\hat{u} = \left(-\frac{3}{5}, \frac{4}{5}\right)$$

Therefore,

$$D_{\hat{u}} f = (1, 2) \cdot \left(-\frac{3}{5}, \frac{4}{5}\right)$$

$$D_{\hat{u}} f = -\frac{3}{5} + \frac{8}{5} = 1$$

$$\boxed{1}$$

Q4.20.1 | NAT | 2020 | 2M | Ans: 3 | ● ○ ○

Given line integral,

$$\int_C [(y+z)\,dx + (x+z)\,dy + (x+y)\,dz]$$

The line segment from $(0,0,0)$ to $(1,1,1)$ can be parameterized as

$$x = t, \quad y = t, \quad z = t$$

where

$$0 \le t \le 1$$

Hence,

$$dx = dy = dz = dt$$

Substituting into the integral,

$$\int_0^1 [(t+t)dt + (t+t)dt + (t+t)dt]$$

$$= \int_0^1 (2t + 2t + 2t)\,dt$$

$$= \int_0^1 6t\,dt$$

$$= 6\left[\frac{t^2}{2}\right]_0^1$$

$$= 3$$

$$\boxed{3}$$

Key Points

- Convert the curve into parametric form before evaluating line integrals.
- Replace dx, dy, and dz using the parameter.
- Limits of parameter are obtained from end points of the curve.

Q4.18.1 | NAT | 2018 | 2M | Ans: 0.67 | ● ○ ○

Given vector field,

$$\vec{F}(r) = x^2\hat{i} + y^2\hat{j}$$

The straight line joining $(0,0)$ to $(1,1)$ can be parameterized as

$$x = t, \qquad y = t$$

where

$$0 \le t \le 1$$

Hence,

$$dx = dt, \qquad dy = dt$$

The line integral is

$$\int_C \vec{F} \cdot d\vec{r} = \int_C (x^2\,dx + y^2\,dy)$$

Substituting the parametric values,

$$= \int_0^1 (t^2\,dt + t^2\,dt)$$

$$= 2\int_0^1 t^2\,dt$$

$$= 2\left[\frac{t^3}{3}\right]_0^1$$

$$= \frac{2}{3}$$

$$\frac{2}{3} = 0.67$$

Therefore, the correct answer is

$$\boxed{0.67}$$

Q4.15.1 NAT 2015 2M Ans: -5.7 ● ○ ○

Given scalar field,

$$u(x, y, z) = x^2 - 3yz$$

Directional derivative in the direction of vector

$$\vec{a} = \hat{i} + \hat{j} - 2\hat{k}$$

is given by

$$D_{\hat{a}}u = \nabla u \cdot \hat{a}$$

First, calculate gradient of u:

$$\nabla u = \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z}\right) = (2x, -3z, -3y)$$

At point $(2, -1, 4)$,

$$\nabla u = (4, -12, 3)$$

Magnitude of direction vector is

$$|\vec{a}| = \sqrt{1^2 + 1^2 + (-2)^2} = \sqrt{6}$$

Hence, unit vector along the direction is

$$\hat{a} = \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}}\right)$$

Therefore,

$$D_{\hat{a}}u = (4, -12, 3) \cdot \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}} \right)$$
$$= \frac{4 - 12 - 6}{\sqrt{6}}$$
$$= -\frac{14}{\sqrt{6}} = -5.71$$

$$\boxed{-5.71}$$

Q4.12.1 MCQ 2012 2M Ans: A ● ○ ○

For the parallelogram, the adjacent sides are represented by

$$\overrightarrow{OP} = a\hat{i} + b\hat{j}$$

and

$$\overrightarrow{OR} = c\hat{i} + d\hat{j}$$

The area of a parallelogram formed by two vectors is equal to the magnitude of their determinant:

$$\text{Area} = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

Therefore,

$$\text{Area} = ad - bc$$

$$\boxed{\text{(a) } ad - bc}$$

Key Points

- Area of a parallelogram formed by vectors $\vec{u} = (u_1, u_2)$ and $\vec{v} = (v_1, v_2)$ is $|u_1 v_2 - u_2 v_1|$.
- Determinant method is the fastest way to find vector area in two dimensions.

Mistakes to be avoided

- Do not use the dot product formula $ac + bd$ for area calculation.
- Keep the determinant order correct while expanding.

Q4.11.1 MCQ 2011 1M Ans: A ● ○ ○

For two vectors $\vec{a}$ and $\vec{b}$,

$$|\vec{a} \times \vec{b}| = ab \sin\theta$$

Squaring both sides,

$$|\vec{a} \times \vec{b}|^2 = a^2 b^2 \sin^2\theta$$

Using the identity

$$\sin^2\theta = 1 - \cos^2\theta$$

we get

$$|\vec{a} \times \vec{b}|^2 = a^2 b^2 (1 - \cos^2\theta)$$

Since

$$\vec{a} \cdot \vec{b} = ab\cos\theta$$

therefore,

$$(\vec{a} \cdot \vec{b})^2 = a^2 b^2 \cos^2 \theta$$

Substituting,

$$|\vec{a} \times \vec{b}|^2 = a^2 b^2 - (\vec{a} \cdot \vec{b})^2$$

$$\boxed{\text{(a) } a^2 b^2 - (\vec{a} \cdot \vec{b})^2}$$

Q4.09.1 MCQ | 2009 | 2M | Ans: B | ● ○ ○

For the scalar function

$$f(x, y, z) = x^2 + 3y^2 + 2z^2$$

the gradient is

$$\nabla f = \left(\frac{\partial f}{\partial x}\right) \hat{i} + \left(\frac{\partial f}{\partial y}\right) \hat{j} + \left(\frac{\partial f}{\partial z}\right) \hat{k}$$

Hence,

$$\nabla f = 2x\hat{i} + 6y\hat{j} + 4z\hat{k}$$

At the point $P(1, 2, -1)$,

$$\nabla f = (2)\hat{i} + (12)\hat{j} + (-4)\hat{k}$$

The given direction vector is

$$\hat{i} - \hat{j} + 2\hat{k}$$

Its magnitude is

$$\sqrt{1^2 + (-1)^2 + 2^2} = \sqrt{6}$$

Therefore, the unit vector is

$$\frac{1}{\sqrt{6}}(\hat{i} - \hat{j} + 2\hat{k})$$

Directional derivative:

$$D_{\hat{u}} f = \nabla f \cdot \hat{u}$$

$$D_{\hat{u}} f = (2, 12, -4) \cdot \left(\frac{1}{\sqrt{6}}(1, -1, 2)\right)$$

$$D_{\hat{u}} f = \frac{1}{\sqrt{6}}(2 - 12 - 8) = \frac{-18}{\sqrt{6}} = -3\sqrt{6}$$

$$\boxed{\text{(b) } -3\sqrt{6}}$$

Q4.09.1 MCQ | 2009 | 1M | Ans: B | ● ○ ○

For the scalar function

$$f(x, y, z) = x^2 + 3y^2 + 2z^2$$

the gradient is defined as

$$\nabla f = \left(\frac{\partial f}{\partial x}\right) \hat{i} + \left(\frac{\partial f}{\partial y}\right) \hat{j} + \left(\frac{\partial f}{\partial z}\right) \hat{k}$$

Calculating the partial derivatives,

$$\frac{\partial f}{\partial x} = 2x, \qquad \frac{\partial f}{\partial y} = 6y, \qquad \frac{\partial f}{\partial z} = 4z$$

Hence,

$$\nabla f = 2x\hat{i} + 6y\hat{j} + 4z\hat{k}$$

At the point $P(1, 2, -1)$,

$$\nabla f = 2(1)\hat{i} + 6(2)\hat{j} + 4(-1)\hat{k}$$

$$\nabla f = 2\hat{i} + 12\hat{j} - 4\hat{k}$$

$$\boxed{\text{(b) } 2\hat{i} + 12\hat{j} - 4\hat{k}}$$

Q4.06.1 MCQ | 2006 | 2M | Ans: C | ● ○ ○

For the function

$$f(x, y, z) = 2x^2 + 3y^2 + z^2$$

the gradient is

$$\nabla f = \left(\frac{\partial f}{\partial x}\right)\hat{i} + \left(\frac{\partial f}{\partial y}\right)\hat{j} + \left(\frac{\partial f}{\partial z}\right)\hat{k}$$

Thus,

$$\nabla f = 4x\hat{i} + 6y\hat{j} + 2z\hat{k}$$

At the point $P(2, 1, 3)$,

$$\nabla f = 8\hat{i} + 6\hat{j} + 6\hat{k}$$

The given direction vector is

$$\vec{a} = \hat{i} - 2\hat{k}$$

Its magnitude is

$$|\vec{a}| = \sqrt{1^2 + (-2)^2} = \sqrt{5}$$

Hence, the unit vector is

$$\hat{u} = \frac{1}{\sqrt{5}}(\hat{i} - 2\hat{k})$$

Directional derivative:

$$D_{\hat{u}} f = \nabla f \cdot \hat{u}$$

$$D_{\hat{u}} f = (8, 6, 6) \cdot \left(\frac{1}{\sqrt{5}}(1, 0, -2)\right)$$

$$D_{\hat{u}} f = \frac{1}{\sqrt{5}}(8 - 12) = \frac{-4}{\sqrt{5}} \approx -1.789$$

$$\boxed{(c) - 1.789}$$

Q4.05.1 MCQ | 2005 | 2M | Ans: C | ● ○ ○

Given,

$$\oint_C (xy\,dy - y^2\,dx)$$

Here,

$$M = -y^2, \qquad N = xy$$

Using Green's theorem,

$$\oint_C (M\,dx + N\,dy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy$$

$$\frac{\partial N}{\partial x} = y, \qquad \frac{\partial M}{\partial y} = -2y$$

Hence,

$$\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = y - (-2y) = 3y$$

The region is the unit square:

$$0 \le x \le 1, \qquad 0 \le y \le 1$$

Therefore,

$$\oint_C (xy\,dy - y^2\,dx) = \int_0^1 \int_0^1 3y\,dx\,dy$$

$$= \int_0^1 3y(1)\,dy = 3\int_0^1 y\,dy$$

$$= 3\left[\frac{y^2}{2}\right]_0^1 = \frac{3}{2}$$

$$\boxed{\text{(c) } \frac{3}{2}}$$

Q4.03.1 MCQ 2003 1M Ans: A ● ○ ○

The plane OQR passes through the origin and the points

$$Q(1,3,4), \qquad R(2,1,-2)$$

A normal vector to the plane is obtained using the cross product:

$$\vec{n} = \overrightarrow{OQ} \times \overrightarrow{OR}$$

Method 1

$$\vec{n} = \overrightarrow{OQ} \times \overrightarrow{OR} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 3 & 4 \\ 2 & 1 & -2 \end{vmatrix}$$

$$\vec{n} = (-10)\hat{i} + 10\hat{j} - 5\hat{k}.$$

Now,

$$\vec{n} = -5(2\hat{i} - 2\hat{j} + \hat{k}).$$

Since the direction of the normal alone determines a plane, any nonzero scalar multiple of a normal vector is also a valid normal vector. Therefore, we may choose the simpler equivalent normal vector

$$\vec{n}_1 = 2\hat{i} - 2\hat{j} + \hat{k}.$$

(This is mathematically equivalent to first writing the plane equation with $\vec{n}$ and then dividing the entire equation by -5.)

Hence, the equation of the plane is

$$\vec{n}_1 \cdot (x\hat{i} + y\hat{j} + z\hat{k}) = 0$$

$$2x - 2y + z = 0.$$

The distance of point

$$P(3,-2,-1)$$

from the plane is

$$d = \frac{|2(3) - 2(-2) + (-1)|}{\sqrt{2^2 + (-2)^2 + 1^2}}$$
$$= \frac{|6 + 4 - 1|}{\sqrt{9}} = \frac{9}{3} = 3.$$

Method 2

Using scalar triple product,

$$d = \frac{|\overrightarrow{OP} \cdot (\overrightarrow{OQ} \times \overrightarrow{OR})|}{|\overrightarrow{OQ} \times \overrightarrow{OR}|}$$

$$\overrightarrow{OP} = (3, -2, -1)$$

$$\overrightarrow{OQ} \times \overrightarrow{OR} = (-10, 10, -5)$$

Thus,

$$d = \frac{|(3, -2, -1) \cdot (-10, 10, -5)|}{\sqrt{(-10)^2 + 10^2 + (-5)^2}}$$
$$= \frac{|-30 - 20 + 5|}{\sqrt{225}} = \frac{45}{15} = 3$$

$$\boxed{\text{(a) } 3}$$

Key Points

- Cross product of two vectors lying in a plane gives a normal vector to the plane.
- Distance of point (x_1, y_1, z_1) from plane $Ax + By + Cz + D = 0$ is

$$\frac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}}$$

Q4.26.1 MCQ 2026 1M Ans: B ● ○ ○

Given,

$$T = 2x^2 + 3xy + y^2$$

The directional derivative vector is obtained using the gradient of T:

$$\nabla T = \frac{\partial T}{\partial x}\hat{\imath} + \frac{\partial T}{\partial y}\hat{\jmath}$$

Now,

$$\frac{\partial T}{\partial x} = 4x + 3y$$

$$\frac{\partial T}{\partial y} = 3x + 2y$$

At the point $(x, y) = (2, 2)$:

$$\frac{\partial T}{\partial x} = 4(2) + 3(2) = 8 + 6 = 14$$

$$\frac{\partial T}{\partial y} = 3(2) + 2(2) = 6 + 4 = 10$$

Hence,

$$\nabla T = 14\hat{\imath} + 10\hat{\jmath}$$

$$\boxed{\text{(b) } 14\hat{\imath} + 10\hat{\jmath}}.$$

Q4.26.2 MCQ 2026 2M Ans: C ● ○ ○

Given velocity field,

$$\vec{V} = (x + y)\hat{\imath} + (y + z)\hat{\jmath} + (z + x)\hat{k}$$

For incompressibility, check divergence:

$$\nabla \cdot \vec{V} = \frac{\partial}{\partial x}(x + y) + \frac{\partial}{\partial y}(y + z) + \frac{\partial}{\partial z}(z + x)$$

$$\nabla \cdot \vec{V} = 1 + 1 + 1 = 3$$

Since,

$$\nabla \cdot \vec{V} \neq 0$$

the flow is compressible.
Now, check rotation using curl:

$$\nabla \times \vec{V} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x + y & y + z & z + x \end{vmatrix}$$

$$\nabla \times \vec{V} = (0 - 1)\hat{\imath} + (0 - 1)\hat{\jmath} + (0 - 1)\hat{k}$$

$$\nabla \times \vec{V} = -\hat{\imath} - \hat{\jmath} - \hat{k} \neq 0$$

Hence, the flow is rotational.

$$\boxed{\text{(c) The flow is compressible and rotational}}$$

Key Points

- A flow is incompressible if:
$$\nabla \cdot \vec{V} = 0$$
- A flow is irrotational if:
$$\nabla \times \vec{V} = 0$$
- Non-zero curl indicates rotational motion of fluid particles.

Q4.25.1 | MCQ | 2025 | 2M | Ans: A | ● ○ ○

Given vector field,

$$\vec{v} = ax\hat{i} - by\hat{j}$$

The divergence of a vector field is:

$$\nabla \cdot \vec{v} = \frac{\partial}{\partial x}(ax) + \frac{\partial}{\partial y}(-by)$$

$$\nabla \cdot \vec{v} = a - b$$

For zero divergence,

$$a - b = 0$$

$$a = b$$

Hence, the correct condition is

$$\boxed{a - b = 0}$$

$$\boxed{\text{(a) } 0}$$

Q4.24.1 | NAT | 2024 | 2M | Ans: 4.93 | ● ● ○

Given,

$$\vec{F}(r) = x\hat{i} + y\hat{j} + z\hat{k}$$

and the curve

$$\vec{r}(t) = \cos t\,\hat{i} + \sin t\,\hat{j} + t\,\hat{k}, \qquad 0 \le t \le \pi$$

From the path equation,

$$x = \cos t, \qquad y = \sin t, \qquad z = t$$

Hence,

$$\vec{F}(r(t)) = \cos t\,\hat{i} + \sin t\,\hat{j} + t\,\hat{k}$$

Now,

$$\frac{d\vec{r}}{dt} = -\sin t\,\hat{i} + \cos t\,\hat{j} + \hat{k}$$

Therefore,

$$\vec{F} \cdot d\vec{r} = \vec{F}(r(t)) \cdot \frac{d\vec{r}}{dt}\,dt$$

$$= (\cos t)(-\sin t) + (\sin t)(\cos t) + (t)(1)\,dt$$

$$= t\,dt$$

Thus,

$$\int_C \vec{F}(r) \cdot d\vec{r} = \int_0^{\pi} t\,dt$$

$$= \left[\frac{t^2}{2}\right]_0^{\pi} = \frac{\pi^2}{2}$$

$$\frac{\pi^2}{2} \approx 4.93$$

$$\boxed{4.93}$$

Q4.23.1 MCQ 2023 1M Ans: D ● ○ ○

Given,

$$\hat{e}_r = \cos\theta\,\hat{i} + \sin\theta\,\hat{j}$$

$$\hat{e}_\theta = -\sin\theta\,\hat{i} + \cos\theta\,\hat{j}$$

We need to find

$$\frac{\partial(\hat{e}_r + \hat{e}_\theta)}{\partial\theta}$$

Using differentiation,

$$\frac{\partial\hat{e}_r}{\partial\theta} = -\sin\theta\,\hat{i} + \cos\theta\,\hat{j} = \hat{e}_\theta$$

Also,

$$\frac{\partial\hat{e}_\theta}{\partial\theta} = -\cos\theta\,\hat{i} - \sin\theta\,\hat{j} = -\hat{e}_r$$

Therefore,

$$\frac{\partial(\hat{e}_r + \hat{e}_\theta)}{\partial\theta} = \frac{\partial\hat{e}_r}{\partial\theta} + \frac{\partial\hat{e}_\theta}{\partial\theta}$$

$$= \hat{e}_\theta - \hat{e}_r$$

$$\boxed{\text{(d) } -\hat{e}_r + \hat{e}_\theta}$$

Q4.22.1 NAT 2022 2M Ans: 192 ● ○ ○

Given vector field,

$$\vec{F} = \frac{2x\hat{i} + 3y\hat{j} + 4z\hat{k}}{4\pi}$$

We need to evaluate

$$\iint_S \vec{F} \cdot \hat{n}\, dA$$

Using Divergence Theorem,

$$\iint_S \vec{F} \cdot \hat{n}\, dA = \iiint_V (\nabla \cdot \vec{F})\, dV$$

Now,

$$\nabla \cdot \vec{F} = \frac{1}{4\pi}\left(\frac{\partial(2x)}{\partial x} + \frac{\partial(3y)}{\partial y} + \frac{\partial(4z)}{\partial z}\right)$$

$$\nabla \cdot \vec{F} = \frac{1}{4\pi}(2 + 3 + 4) = \frac{9}{4\pi}$$

Volume of sphere of radius 4:

$$V = \frac{4}{3}\pi(4)^3 = \frac{256\pi}{3}$$

Hence,

$$\iint_S \vec{F} \cdot \hat{n}\, dA = \frac{9}{4\pi} \times \frac{256\pi}{3}$$

$$= 192$$

$$\boxed{192}$$

Q4.26.1 NAT | 2026 | 1M | Ans: 1

Given,

$$\int_C (2x\,dx + 2y\,dy + 2z\,dz)$$

Observe that,

$$2x\,dx + 2y\,dy + 2z\,dz = d(x^2 + y^2 + z^2)$$

Hence, the integral becomes

$$\int_C d(x^2 + y^2 + z^2)$$

Using the fundamental theorem for line integrals,

$$\int_C d\phi = \phi(\text{end}) - \phi(\text{start})$$

where

$$\phi = x^2 + y^2 + z^2$$

Start point:

$$(0, 0, 0) \Rightarrow \phi = 0$$

End point:

$$(1, 0, 0) \Rightarrow \phi = 1$$

Therefore,

$$\int_C (2x\,dx + 2y\,dy + 2z\,dz) = 1$$

$$\boxed{1}$$

Key Points

- If the integrand is an exact differential, the line integral is path independent.
- Here,
$$d(x^2 + y^2 + z^2) = 2x\,dx + 2y\,dy + 2z\,dz$$
- Only initial and final points are needed.

Mistakes to be avoided

- Do not unnecessarily parameterize the semicircle.
- Check whether the integrand is an exact differential first.

Q4.25.1 NAT 2025 2M Ans: 2 ● ○ ○

Given,

$$f(x, y, z) = x^2 + 2y^2 + z$$

The directional derivative is given by

$$\nabla f \cdot \hat{u}$$

where $\hat{u}$ is the unit vector along the given direction.
First, compute the gradient:

$$\nabla f = \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} + \frac{\partial f}{\partial z}\hat{k}$$

$$\nabla f = 2x\,\hat{i} + 4y\,\hat{j} + \hat{k}$$

At the point $(1, 1, 1)$,

$$\nabla f = 2\hat{i} + 4\hat{j} + \hat{k}$$

The given direction vector is

$$3\hat{i} + 4\hat{k}$$

Its magnitude is

$$\sqrt{3^2 + 4^2} = 5$$

Hence, the unit vector is

$$\hat{u} = \frac{3}{5}\hat{i} + \frac{4}{5}\hat{k}$$

Therefore,

$$\nabla f \cdot \hat{u} = (2\hat{i} + 4\hat{j} + \hat{k}) \cdot \left(\frac{3}{5}\hat{i} + \frac{4}{5}\hat{k}\right)$$

$$= \frac{6}{5} + \frac{4}{5} = \frac{10}{5} = 2$$

$$\boxed{2}$$

Q4.24.1 MCQ 2024 1M Ans: * ● ● ○

Given vector field,

$$\vec{F}(x, y) = (100x + 100y)\hat{i} + (-50x + 200y)\hat{j}$$

We need to evaluate

$$\oint_C \vec{F} \cdot d\vec{l}$$

where

$$d\vec{l} = dx\,\hat{i} + dy\,\hat{j}$$

Using Green's theorem,

$$\oint_C (P\,dx + Q\,dy) = \iint_A \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$$

Here,

$$P = 100x + 100y, \qquad Q = -50x + 200y$$

Therefore,

$$\frac{\partial Q}{\partial x} = -50$$

$$\frac{\partial P}{\partial y} = 100$$

Hence,

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = -50 - 100 = -150$$

The contour is a circle of radius 2, so area is

$$A = \pi r^2 = \pi(2)^2 = 4\pi$$

Thus,

$$\oint_C \vec{F} \cdot d\vec{l} = (-150)(4\pi)$$

$$= -600\pi$$

$$\boxed{-600\pi}$$

Q4.22.1 MCQ 2022 1M Ans: B ● ○ ○

Using the vector triple product identity,

$$(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}$$

$$\boxed{\text{(b) } (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}}$$

Key Points

- Vector triple product identity:
$$(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}$$
- The result of a vector triple product is always a vector lying in the plane of the vectors involved.
- Remember the pattern:
$$(\vec{A} \times \vec{B}) \times \vec{C} = (\vec{A} \cdot \vec{C})\vec{B} - (\vec{B} \cdot \vec{C})\vec{A}$$

Mistakes to be avoided

- Do not confuse it with $\vec{a} \times (\vec{b} \times \vec{c})$ which has a different formula.
- Cross product is not associative.

Q4.18.1 NAT | 2018 | 1M | Ans: 678.24 | ● ○ ○

Given vector field,

$$\vec{F} = 9x\hat{i} - 2y\hat{j} - z\hat{k}$$

We need to evaluate

$$\iint_S \vec{F} \cdot \hat{n}\, dS$$

over the sphere

$$x^2 + y^2 + z^2 = 9$$

Using Divergence Theorem,

$$\iint_S \vec{F} \cdot \hat{n}\, dS = \iiint_V (\nabla \cdot \vec{F})\, dV$$

Now,

$$\nabla \cdot \vec{F} = \frac{\partial(9x)}{\partial x} + \frac{\partial(-2y)}{\partial y} + \frac{\partial(-z)}{\partial z}$$

$$\nabla \cdot \vec{F} = 9 - 2 - 1 = 6$$

Radius of sphere:

$$r = 3$$

Volume of sphere:

$$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(3)^3 = 36\pi$$

Therefore,

$$\iint_S \vec{F} \cdot \hat{n}\, dS = 6 \times 36\pi$$

$$= 216\pi$$

$$\boxed{678.24}$$

Q4.17.1 MCQ | 2017 | 1M | Ans: D | ● ○ ○

Given,

$$(\vec{A} + \vec{B}) \perp (\vec{A} - \vec{B})$$

For perpendicular vectors, their dot product is zero:

$$(\vec{A} + \vec{B}) \cdot (\vec{A} - \vec{B}) = 0$$

Expanding,

$$\vec{A} \cdot \vec{A} - \vec{A} \cdot \vec{B} + \vec{B} \cdot \vec{A} - \vec{B} \cdot \vec{B} = 0$$

Since,

$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$$

we get

$$|\vec{A}|^2 - |\vec{B}|^2 = 0$$

$$|\vec{A}|^2 = |\vec{B}|^2$$

Hence,

$$|\vec{A}| = |\vec{B}|$$

$$\boxed{\text{(d) The magnitudes of } \vec{A} \text{ and } \vec{B} \text{ are equal}}$$

Q4.14.1 MCQ 2014 2M Ans: C ● ○ ○

Given,

$$\phi = 2x^3y^2z^4$$

The Laplacian operator is

$$\nabla^2\phi = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2}$$

Now,

$$\frac{\partial\phi}{\partial x} = 6x^2y^2z^4$$

$$\frac{\partial^2\phi}{\partial x^2} = 12xy^2z^4$$

Similarly,

$$\frac{\partial\phi}{\partial y} = 4x^3yz^4$$

$$\frac{\partial^2\phi}{\partial y^2} = 4x^3z^4$$

Also,

$$\frac{\partial\phi}{\partial z} = 8x^3y^2z^3$$

$$\frac{\partial^2\phi}{\partial z^2} = 24x^3y^2z^2$$

Therefore,

$$\nabla^2\phi = 12xy^2z^4 + 4x^3z^4 + 24x^3y^2z^2$$

$$\boxed{12xy^2z^4 + 4x^3z^4 + 24x^3y^2z^2}$$

$$\boxed{\text{(c) } 12xy^2z^4 + 4x^3z^4 + 24x^3y^2z^2}.$$

Q4.14.2 MCQ 2014 1M Ans: B ● ○ ○

Given,

$$\phi = 2xz - y^2$$

The directional derivative is maximum in the direction of the gradient vector:

$$\nabla\phi = \frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}$$

Now,

$$\frac{\partial\phi}{\partial x} = 2z$$

$$\frac{\partial\phi}{\partial y} = -2y$$

$$\frac{\partial\phi}{\partial z} = 2x$$

Hence,

$$\nabla\phi = 2z\,\hat{i} - 2y\,\hat{j} + 2x\,\hat{k}$$

At the point $(1, 3, 2)$,

$$\nabla\phi = 4\hat{i} - 6\hat{j} + 2\hat{k}$$

Therefore, the directional derivative becomes maximum in the direction

$$\boxed{4\hat{i} - 6\hat{j} + 2\hat{k}}$$

$$\boxed{\text{(b) } 4\hat{i} - 6\hat{j} + 2\hat{k}}$$

Q4.11.1 MCQ 2011 1M Ans: A ● ○ ○

Given points,

$$A(0, 4, 3), \qquad B(0, 0, 0), \qquad C(3, 0, 4)$$

First, find the vectors:

$$\overrightarrow{AB} = B - A = (0-0)\hat{i} + (0-4)\hat{j} + (0-3)\hat{k}$$

$$\overrightarrow{AB} = -4\hat{j} - 3\hat{k}$$

Also,

$$\overrightarrow{BC} = C - B = 3\hat{i} + 0\hat{j} + 4\hat{k}$$

A vector perpendicular to both is obtained using cross product:

$$\overrightarrow{AB} \times \overrightarrow{BC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & -4 & -3 \\ 3 & 0 & 4 \end{vmatrix}$$

$$= (-16)\hat{i} - (9)\hat{j} + (12)\hat{k}$$

$$= -16\hat{i} - 9\hat{j} + 12\hat{k}$$

Multiplying by -1 gives an equivalent perpendicular vector:

$$16\hat{i} + 9\hat{j} - 12\hat{k}$$

$$\boxed{\text{(a) } 16\hat{i} + 9\hat{j} - 12\hat{k}}$$

Q4.11.2 MCQ 2011 1M Ans: A ● ● ○

Given line integral,

$$\int_{P_1}^{P_2} (y\,dx + x\,dy)$$

Observe that,

$$d(xy) = x\,dy + y\,dx$$

Hence,

$$y\,dx + x\,dy = d(xy)$$

Therefore,

$$\int_{P_1}^{P_2} (y\,dx + x\,dy) = \int_{P_1}^{P_2} d(xy)$$

Using the fundamental theorem for line integrals,

$$\int_{P_1}^{P_2} d(xy) = (xy)\Big|_{P_1}^{P_2}$$

$$= x_2y_2 - x_1y_1$$

Thus,

$$\boxed{\text{(a) } x_2y_2 - x_1y_1}$$

Q4.08.1 MCQ 2008 1M Ans: D ● ○ ○

Let,

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

Using the Divergence Theorem,

$$\iint_S \vec{r} \cdot d\vec{S} = \iiint_V (\nabla \cdot \vec{r})\,dV$$

Now,

$$\nabla \cdot \vec{r} = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z}$$

$$\nabla \cdot \vec{r} = 1 + 1 + 1 = 3$$

Therefore,

$$\iint_S \vec{r} \cdot d\vec{S} = \iiint_V 3\, dV$$

$$= 3V$$

$$\boxed{\text{(d) 3V}}$$

Q4.07.1 MCQ 2007 2M Ans: A ● ○ ○

Given vector field,

$$\vec{V}(x, y, z) = -(x \cos xy + y)\hat{\imath} + (y \cos xy)\hat{\jmath} + \left[\sin z^2 + x^2 + y^2\right] \hat{k}$$

The divergence is

$$\nabla \cdot \vec{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$$

Now,

$$V_x = -(x \cos xy + y)$$

$$\frac{\partial V_x}{\partial x} = -(\cos xy - xy \sin xy)$$

Also,

$$V_y = y \cos xy$$

$$\frac{\partial V_y}{\partial y} = \cos xy - xy \sin xy$$

Further,

$$V_z = \sin z^2 + x^2 + y^2$$

$$\frac{\partial V_z}{\partial z} = 2z \cos z^2$$

Adding,

$$\nabla \cdot \vec{V} = -(\cos xy - xy \sin xy) + (\cos xy - xy \sin xy) + 2z \cos z^2$$

$$\nabla \cdot \vec{V} = 2z \cos z^2$$

$$\boxed{\text{(a) } 2z \cos z^2}$$

DEBUG INFO

Vector Calculus

Engineering Mathematics

Total Questions : 145

MCQ : 101

MSQ : 4

NAT : 40